

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T10:11:13

Source: data/json/noise_experiments/v2/noise2i_match8_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_match8/noise2i_match8_game_003_neutral_rep03.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_match8
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 3
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 0
coplayer_history_window: 0
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 3
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$5; payoffs Agent_1=\$7, Agent_2=\$4;

noise/visible: sent visible=\$3.64, received visible=\$10.92, returned visible=\$4.58

Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 7.0,

"Agent_2": 4.0

}

Visible balances after round: {

"Agent_1": 6.58,

"Agent_2": 5.92

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:06:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

I'll send **\$3** to the receiver.

My reasoning: This is the first round, so I want to signal trust and cooperation while managing risk. By sending \$3, it will be multiplied to \$9 for the receiver. If they reciprocate fairly, we can both benefit. This moderate amount allows us to build trust while limiting my downside risk.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- I keep: \$5 - sent + returned
- There may be communication noise

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver is fair and cooperative, they might return half (\$7.50), giving me \$2.50 net gain and them \$7.50.

However, there's risk:

- The receiver might not return anything (I'd lose what I sent)
- There's communication noise, so even if they try to be fair, the amounts might be distorted

Common strategies:

1. Send nothing (safe but no potential gain)
2. Send everything (maximizes total welfare but risky)
3. Send a moderate amount (balances risk and potential gain)

Since this is round 1 and I have no information about the receiver's behavior, I should probably send a moderate amount to signal trust while limiting risk. Sending \$3 seems reasonable:

- Shows willingness to cooperate
- If returned proportionally (half of the \$9 they receive = \$4.50), I'd end up with \$5 - \$3 + \$4.50 = \$6.50
- If they keep everything, I lose \$3 but still have \$2
- It's a middle ground that allows for building trust

Let me send \$3.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 458,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:06:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
```

```
"round": 1,  
"task": "game",  
"task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.64, that is 72.8% of their total endowment. You received \$10.92. How much do you return to The sender? (0-10.92)

Assistant output:

Looking at this situation:

- The sender showed strong trust by sending \$3.64 (72.8% of their endowment)
- I received \$10.92 (3x the amount sent)
- The sender currently has \$1.36 remaining

This is a multi-round game, so building trust is important. The sender demonstrated significant trust, and I want to reciprocate that while ensuring we both benefit from the cooperation.

If I return approximately \$5.00:

- The sender would have: $\$1.36 + \$5.00 = \$6.36$ (a gain of \$1.36 from their initial \$5)
- I would have: $\$10.92 - \$5.00 = \$5.92$ (a gain of \$5.92)
- Both parties end up better off than if no trust had been shown

This demonstrates reciprocity while maintaining an incentive structure for future cooperation.

```
{"return": 5.00}
```

Reasoning:

Let me think through this game theory situation carefully.

I'm the RECEIVER in round 1.

- The sender had \$5
- The sender sent \$3.64 (which is 72.8% of their endowment)
- I received \$10.92 (which is $\$3.64 \times 3$)

- I need to decide how much to return (between \$0 and \$10.92)

Key considerations:

1. The sender showed significant trust by sending 72.8% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds
3. There's communication noise, so amounts may differ slightly
4. I want to be fair and reciprocate the trust shown

Let me think about what would be a fair return:

- Sender currently has: \$5 - \$3.64 = \$1.36
- I currently have: \$10.92
- Total pool created: \$1.36 + \$10.92 = \$12.28

A reasonable approach could be:

1. Equal split of total: $\$12.28 / 2 = \6.14 each
 - Sender would need: $\$6.14 - \$1.36 = \$4.78$ returned
 - I would keep: $\$10.92 - \$4.78 = \$6.14$
2. Proportional return based on their trust (they sent 72.8%):
 - They showed high trust, so I should return a substantial amount
 - Maybe return around 40-50% of what I received?
 - 40% of \$10.92 = \$4.37
 - 50% of \$10.92 = \$5.46
3. Ensure sender ends up better off than if they sent nothing:
 - If they sent \$0, they'd have \$5
 - Returning \$5.00 would give them \$6.36, which meaningfully rewards their trust
 - This also leaves me with \$5.92, creating a balanced outcome that encourages continued cooperation in future rounds

I'll return \$5.00 as it demonstrates strong reciprocity while still leaving me with a reasonable amount....

```
Usage: {  
  "input_tokens": 394,  
  "output_tokens": 1129,  
  "reasoning_tokens": 0  
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3.64; received \$10.92; Agent_1 returned \$1.3; payoffs Agent_2=\$2.66,
Agent_1=\$9.62; noise/visible: sent visible=\$0.84, received visible=\$2.52, returned visible=\$1.06
Action validation: {"returned": {"amount": 1.3, "clamped": false, "max": 10.92, "min": 0, "raw": 1.3}, "sent": {"amount": 3.64, "clamped": false, "max": 5, "min": 0, "raw": 3.64}}

Balances after round: {

"Agent_1": 16.619999999999997,
"Agent_2": 6.66

}

Visible balances after round: {

"Agent_1": 7.8,
"Agent_2": 8.34

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:06:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {

```

"agents": [
  "Agent_2",
  "Agent_1"
],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.64, that is 72.8% of their total endowment. You received \$10.92. How much do you return to The sender? (0-10.92)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed strong trust by sending \$3.64 (72.8% of their endowment)
- I received \$10.92 (3x the amount sent)
- The sender currently has \$1.36 remaining

This is a multi-round game, so building trust is important. The sender demonstrated significant trust, and I want to reciprocate that while ensuring we both benefit from the cooperation.

If I return approximately \$5.00:

- The sender would have: $\$1.36 + \$5.00 = \$6.36$ (a gain of \$1.36 from their initial \$5)
- I would have: $\$10.92 - \$5.00 = \$5.92$ (a gain of \$5.92)
- Both parties end up better off than if no trust had been shown

This demonstrates reciprocity while maintaining an incentive structure for future cooperation.

```
{"return": 5.00}
```

reasoning:

Let me think through this game theory situation carefully.

I'm the RECEIVER in round 1.

- The sender had \$5
- The sender sent \$3.64 (which is 72.8% of their endowment)
- I received \$10.92 (which is $\$3.64 \times 3$)
- I need to decide how much to return (between \$0 and \$10.92)

Key considerations:

1. The sender showed significant trust by sending 72.8% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds
3. There's communication noise, so amounts may differ slightly
4. I want to be fair and reciprocate the trust shown

Let me think about what would be a fair return:

- Sender currently has: $\$5 - \$3.64 = \$1.36$
- I currently have: \$10.92
- Total pool created: $\$1.36 + \$10.92 = \$12.28$

A reasonable approach could be:

1. Equal split of total: $\$12.28 / 2 = \6.14 each
 - Sender would need: $\$6.14 - \$1.36 = \$4.78$ returned
 - I would keep: $\$10.92 - \$4.78 = \$6.14$
2. Proportional return based on their trust (they sent 72.8%):
 - They showed high trust, so I should return a substantial amount
 - Maybe return around 40-50% of what I received?
 - 40% of \$10.92 = \$4.37
 - 50% of \$10.92 = \$5.46
3. Ensure sender ends up better off than if they sent nothing:
 - If they sent \$0, they'd have \$5
 - Returning \$5.00 would give them \$6.36, which meaningfully rewards their trust
 - This also leaves me with \$5.92, creating a balanced outcome that encourages continued cooperation in future rounds

I'll return \$5.00 as it demonstrates strong reciprocity while still leaving me with a reasonable amount....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1129,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.92.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the previous round, the sender showed high trust by sending \$3.64 (72.8% of their endowment), and I reciprocated by returning a fair amount.

Now as the sender, I want to maintain that cooperative dynamic. In multi-round games, establishing a pattern of trust and reciprocity tends to benefit everyone.

I'll send a similar proportion to demonstrate I'm willing to trust at the same level:

```
{"send": 3.64}
```


Reasoning:

Now I'm the SENDER in Round 2. Let me think about this situation:

Previous Round 1:

- I was the receiver
- Sender sent me \$3.64 (72.8% of their endowment)
- I received \$10.92
- I returned \$5.00
- Sender ended with \$6.36, I ended with \$5.92

Now in Round 2:

- I start with a fresh \$5 endowment
- My total earnings across all rounds are \$5.92 (from Round 1)
- I need to decide how much to send (0-5)

The sender in Round 1 showed high trust by sending 72.8%. I want to reciprocate that trust level and maintain cooperation in this multi-round game. If I send a similar proportion, that would be around \$3.50-\$3.75.

Let me consider:

- If I send \$3.50: Receiver gets \$10.50
- If I send \$3.64: Receiver gets \$10.92 (same as what I received)
- If I send \$3.75: Receiver gets \$11.25

Sending \$3.64 seems like a good reciprocal move - matching what the previous sender sent to me. This shows I'm willing to trust at the same level I was trusted.

Actually, let me reconsider. The person who sent me \$3.64 in Round 1 might or might not be the same person I'm sending to now. The instructions say "You will play different roles across multiple rounds" but don't specify if it's with the same or different partners.

Regardless, building a reputation for trustworthiness in this multi-round game seems beneficial. By sending a similar amount to what I received, I'm:

1. Showing I'm trustworthy and willing to cooperate
2. Potentially building reciprocity for when I'm a receiver again

...

```
Usage: {
  "input_tokens": 658,
  "output_tokens": 572,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:07:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send **\$3** to the receiver.

My reasoning: This is the first round, so I want to signal trust and cooperation while managing risk. By sending \$3, it will be multiplied to \$9 for the receiver. If they reciprocate fairly, we can both benefit. This moderate amount allows us to build trust while limiting my downside risk.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- I keep: \$5 - sent + returned
- There may be communication noise

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver is fair and cooperative, they might return half (\$7.50), giving me \$2.50 net gain and them \$7.50.

However, there's risk:

- The receiver might not return anything (I'd lose what I sent)
- There's communication noise, so even if they try to be fair, the amounts might be distorted

Common strategies:

1. Send nothing (safe but no potential gain)
2. Send everything (maximizes total welfare but risky)
3. Send a moderate amount (balances risk and potential gain)

Since this is round 1 and I have no information about the receiver's behavior, I should probably send a moderate amount to signal trust while limiting risk. Sending \$3 seems reasonable:

- Shows willingness to cooperate
- If returned proportionally (half of the \$9 they receive = \$4.50), I'd end up with $\$5 - \$3 + \$4.50 = \6.50
- If they keep everything, I lose \$3 but still have \$2
- It's a middle ground that allows for building trust

Let me send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 458,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.58.

This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to The sender? (0-2.52)

Assistant output:

Looking at this situation:

- The sender sent me \$0.84, which became \$2.52 after the 3x multiplier
- This is a relatively small amount, suggesting either caution or they may have had a poor experience

I want to reciprocate fairly and encourage continued cooperation. Even though the amount sent was small, I'll return a reasonable portion to show I'm trustworthy.

I'll return ****\$1.30**** to the sender.

This means:

- The sender gains overall ($\$5 - \$0.84 + \$1.30 = \5.46)
- I keep \$1.22 ($\$2.52 - \1.30)
- Both of us profit from the exchange

```
'''json
{"return": 1.30}
'''
```

Reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.84, which was multiplied by 3x to give me \$2.52
3. My total earnings across all rounds are \$6.58

Wait, let me calculate what happened in Round 1:

- I was the sender and sent \$3
- That left me with $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- I need to figure out what they returned to me

If my total earnings are \$6.58, and I had \$2 remaining after sending, then I must have received $\$6.58 - \$2 = \$4.58$ back from the receiver in Round 1.

That means:

- The receiver got \$9
- They returned \$4.58 to me
- They kept $\$9 - \$4.58 = \$4.42$
- I ended with $\$2 + \$4.58 = \$6.58$?

So the receiver in Round 1 returned about 51% of what they received, which is pretty fair/cooperative.

Now in Round 2:

- The sender sent me \$0.84
- I received \$2.52 (after 3x multiplication)
- The note mentions there may be communication noise - amounts I see may differ from what was actually sent

The sender only sent \$0.84, which is quite low. This could mean:

1. They're being cautious/conservative
2. There might be noise in the communication
3. They had a bad experience in their previous round

Given that the previous receiver returned about 51% to me, I should probably reciprocate with a similar level of cooperation. If I return around half or slightly more than half of what I received, that would be fair.

$$\$2.52 \times 0.5 = \$1.26$$

I'll return \$1.30 to show I'm cooperative and willing to reciprocate trust, even though the sender only sent a small amount.

Returning \$1.30 means they get...

```
Usage: {
  "input_tokens": 522,
  "output_tokens": 808,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$4; received \$12; Agent_2 returned \$0.4; payoffs Agent_1=\$1.4, Agent_2=\$11.6;
noise/visible: sent visible=\$0.33, received visible=\$0.99, returned visible=\$1.24
Action validation: {"returned": {"amount": 0.4, "clamped": false, "max": 12.0, "min": 0, "raw": 0.4}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 18.019999999999996,
"Agent_2": 18.259999999999998

}

Visible balances after round: {

"Agent_1": 10.04,
"Agent_2": 8.93

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:07:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",

```

    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send **\$3** to the receiver.

My reasoning: This is the first round, so I want to signal trust and cooperation while managing risk. By sending \$3, it will be multiplied to \$9 for the receiver. If they reciprocate fairly, we can both benefit. This moderate amount allows us to build trust while limiting my downside risk.

```

'''json
{"send": 3}
'''

```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return

- I keep: \$5 - sent + returned
- There may be communication noise

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver is fair and cooperative, they might return half (\$7.50), giving me \$2.50 net gain and them \$7.50.

However, there's risk:

- The receiver might not return anything (I'd lose what I sent)
- There's communication noise, so even if they try to be fair, the amounts might be distorted

Common strategies:

1. Send nothing (safe but no potential gain)
2. Send everything (maximizes total welfare but risky)
3. Send a moderate amount (balances risk and potential gain)

Since this is round 1 and I have no information about the receiver's behavior, I should probably send a moderate amount to signal trust while limiting risk. Sending \$3 seems reasonable:

- Shows willingness to cooperate
- If returned proportionally (half of the \$9 they receive = \$4.50), I'd end up with \$5 - \$3 + \$4.50 = \$6.50
- If they keep everything, I lose \$3 but still have \$2
- It's a middle ground that allows for building trust

Let me send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 458,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.58.

This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to The sender? (0-2.52)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.84, which became \$2.52 after the 3x multiplier
- This is a relatively small amount, suggesting either caution or they may have had a poor experience

I want to reciprocate fairly and encourage continued cooperation. Even though the amount sent was small, I'll return a reasonable portion to show I'm trustworthy.

I'll return **\$1.30** to the sender.

This means:

- The sender gains overall (\$5 - \$0.84 + \$1.30 = \$5.46)
- I keep \$1.22 (\$2.52 - \$1.30)
- Both of us profit from the exchange

```
'''json
{"return": 1.30}
'''
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.84, which was multiplied by 3x to give me \$2.52
3. My total earnings across all rounds are \$6.58

Wait, let me calculate what happened in Round 1:

- I was the sender and sent \$3

- That left me with $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- I need to figure out what they returned to me

If my total earnings are \$6.58, and I had \$2 remaining after sending, then I must have received $\$6.58 - \$2 = \$4.58$ back from the receiver in Round 1.

That means:

- The receiver got \$9
- They returned \$4.58 to me
- They kept $\$9 - \$4.58 = \$4.42$
- I ended with $\$2 + \$4.58 = \$6.58$?

So the receiver in Round 1 returned about 51% of what they received, which is pretty fair/cooperative.

Now in Round 2:

- The sender sent me \$0.84
- I received \$2.52 (after 3x multiplication)
- The note mentions there may be communication noise - amounts I see may differ from what was actually sent

The sender only sent \$0.84, which is quite low. This could mean:

1. They're being cautious/conservative
2. There might be noise in the communication
3. They had a bad experience in their previous round

Given that the previous receiver returned about 51% to me, I should probably reciprocate with a similar level of cooperation. If I return around half or slightly more than half of what I received, that would be fair.

$$\$2.52 \times 0.5 = \$1.26$$

I'll return \$1.30 to show I'm cooperative and willing to reciprocate trust, even though the sender only sent a small amount.

Returning \$1.30 means they get...

usage:

```
{
  "input_tokens": 522,
  "output_tokens": 808,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.8.
How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my earnings history:

In Round 1, I sent \$3 (which became \$9 for the receiver), and based on my total earnings, the receiver returned a generous amount - they were cooperative and trustworthy.

In Round 2, I received \$0.84 from a sender and returned \$1.30 out of \$2.52, showing I'm also willing to reciprocate fairly.

Now in Round 3, I want to continue building trust and cooperation. Given the positive reciprocity I've experienced, I'll send **\$4**.

This will give the receiver \$12 to work with. If they continue the cooperative pattern, we should both benefit significantly.

```
'''json
{"send": 4}
'''
```

Reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. My total earnings are now \$7.8 across all rounds.

Let me work backwards:

- Round 1: I sent \$3, so I had $\$5 - \$3 = \$2$ left. The receiver got \$9 and returned some amount to me.
- Round 2: I was the receiver. I got \$2.52 and returned \$1.30, so I kept \$1.22.

So across rounds 1 and 2, my earnings should be:

Round 1: $\$2 + (\text{amount returned from round 1})$

Round 2: \$1.22

Total = $\$2 + \text{amount_returned_R1} + \$1.22 = \$3.22 + \text{amount_returned_R1}$

If total is \$7.8, then:

$\$3.22 + \text{amount_returned_R1} = \7.8

$\text{amount_returned_R1} = \4.58

So in Round 1, the receiver returned \$4.58 to me. That's a very generous return! They received \$9 and returned \$4.58, keeping \$4.42.

This shows the receiver was quite cooperative and fair, returning more than half of what they received.

Now in Round 3, I'm the sender again. Given:

1. The receiver was very generous in Round 1 (returned \$4.58 out of \$9)
2. In Round 2, the sender only sent me \$0.84 (which seems low - they may have had a bad experience or are being cautious)
3. But I returned \$1.30 out of \$2.52, which was a good return rate

I should reciprocate the trust shown by the Round 1 receiver. Since they were generous with me, I should send a good amount. I'll send \$4 this time - a bit more than my initial \$3, showing that I appreciate the cooperation and trust.

...

```
Usage: {  
  "input_tokens": 743,  
  "output_tokens": 637,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:07:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.64, that is 72.8% of their total endowment. You received \$10.92. How much do you return to The sender? (0-10.92)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed strong trust by sending \$3.64 (72.8% of their endowment)
- I received \$10.92 (3x the amount sent)
- The sender currently has \$1.36 remaining

This is a multi-round game, so building trust is important. The sender demonstrated significant trust, and I want to reciprocate that while ensuring we both benefit from the cooperation.

If I return approximately \$5.00:

- The sender would have: $\$1.36 + \$5.00 = \$6.36$ (a gain of \$1.36 from their initial \$5)
- I would have: $\$10.92 - \$5.00 = \$5.92$ (a gain of \$5.92)
- Both parties end up better off than if no trust had been shown

This demonstrates reciprocity while maintaining an incentive structure for future cooperation.

{"return": 5.00}

reasoning:

Let me think through this game theory situation carefully.

I'm the RECEIVER in round 1.

- The sender had \$5
- The sender sent \$3.64 (which is 72.8% of their endowment)
- I received \$10.92 (which is $\$3.64 \times 3$)
- I need to decide how much to return (between \$0 and \$10.92)

Key considerations:

1. The sender showed significant trust by sending 72.8% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds

3. There's communication noise, so amounts may differ slightly
4. I want to be fair and reciprocate the trust shown

Let me think about what would be a fair return:

- Sender currently has: $\$5 - \$3.64 = \$1.36$
- I currently have: $\$10.92$
- Total pool created: $\$1.36 + \$10.92 = \$12.28$

A reasonable approach could be:

1. Equal split of total: $\$12.28 / 2 = \6.14 each
 - Sender would need: $\$6.14 - \$1.36 = \$4.78$ returned
 - I would keep: $\$10.92 - \$4.78 = \$6.14$
2. Proportional return based on their trust (they sent 72.8%):
 - They showed high trust, so I should return a substantial amount
 - Maybe return around 40-50% of what I received?
 - 40% of $\$10.92 = \4.37
 - 50% of $\$10.92 = \5.46
3. Ensure sender ends up better off than if they sent nothing:
 - If they sent $\$0$, they'd have $\$5$
 - Returning $\$5.00$ would give them $\$6.36$, which meaningfully rewards their trust
 - This also leaves me with $\$5.92$, creating a balanced outcome that encourages continued cooperation in future rounds

I'll return $\$5.00$ as it demonstrates strong reciprocity while still leaving me with a reasonable amount....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1129,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are $\$5.92$.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed high trust by sending $\$3.64$ (72.8% of their endowment), and I reciprocated by returning a fair amount.

Now as the sender, I want to maintain that cooperative dynamic. In multi-round games, establishing a pattern of trust and reciprocity tends to benefit everyone.

I'll send a similar proportion to demonstrate I'm willing to trust at the same level:

```
{"send": 3.64}
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this situation:

Previous Round 1:

- I was the receiver
- Sender sent me $\$3.64$ (72.8% of their endowment)
- I received $\$10.92$
- I returned $\$5.00$
- Sender ended with $\$6.36$, I ended with $\$5.92$

Now in Round 2:

- I start with a fresh $\$5$ endowment
- My total earnings across all rounds are $\$5.92$ (from Round 1)
- I need to decide how much to send (0-5)

The sender in Round 1 showed high trust by sending 72.8%. I want to reciprocate that trust level and maintain cooperation in this multi-round game. If I send a similar proportion, that would be around $\$3.50$ - $\$3.75$.

Let me consider:

- If I send \$3.50: Receiver gets \$10.50
- If I send \$3.64: Receiver gets \$10.92 (same as what I received)
- If I send \$3.75: Receiver gets \$11.25

Sending \$3.64 seems like a good reciprocal move - matching what the previous sender sent to me. This shows I'm willing to trust at the same level I was trusted.

Actually, let me reconsider. The person who sent me \$3.64 in Round 1 might or might not be the same person I'm sending to now. The instructions say "You will play different roles across multiple rounds" but don't specify if it's with the same or different partners.

Regardless, building a reputation for trustworthiness in this multi-round game seems beneficial. By sending a similar amount to what I received, I'm:

1. Showing I'm trustworthy and willing to cooperate
2. Potentially building reciprocity for when I'm a receiver again

...

usage:

```
{
  "input_tokens": 658,
  "output_tokens": 572,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.34.

This round, The sender sent you \$0.33, so you received \$0.99. How much do you return to The sender? (0-0.99)

Assistant output:

This is a significant change from Round 1. The sender only sent \$0.33 (6.6% of their endowment) compared to the high trust shown initially.

Given the note about communication noise, it's possible that what I sent in Round 2 appeared different to them, or what they returned appeared different to me, affecting this round's decision.

Even with lower trust, I want to show I'm still willing to cooperate and maintain some reciprocity. I received \$0.99, so I'll return a fair proportion:

```
{"return": 0.40}
```

Reasoning:

This is interesting. The sender only sent \$0.33 (6.6% of their endowment) this round, which is a drastic reduction from the \$3.64 they sent in Round 1.

Let me think about what happened:

- Round 1: Sender sent \$3.64, I returned \$5.00
- Round 2: I sent \$3.64 (as the sender)
- Round 3: Sender sent \$0.33

Wait, there's a note about communication noise. The prompt says "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So what could have happened:

1. In Round 2, I sent \$3.64, but maybe the receiver saw a different amount due to noise
2. The receiver might have thought I sent less or betrayed them somehow
3. Or they might have returned very little
4. Now in Round 3, they're only sending \$0.33

Alternatively:

- Maybe the receiver in Round 2 returned very little, and now as the sender in Round 3, they're being cautious
- Or the noise affected what they saw in previous rounds

Given the communication noise and the significant drop in trust, I have a few options:

1. Return nothing or very little as retaliation
2. Return a fair proportion to try to rebuild trust
3. Return a generous amount to signal I want to re-establish cooperation

Since:

- I received \$0.99 total
- The sender only has \$4.67 remaining (\$5 - \$0.33)
- This is a multi-round game where cooperation benefits everyone

I think I should return a proportion that's fair and shows willingness to cooperate, even if they aren't fully trusting right now. This might be due to noise in communication.

If I return about \$0.40:

```
...
Usage: {
  "input_tokens": 827,
  "output_tokens": 688,
  "reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2; received \$6; Agent_1 returned \$0; payoffs Agent_2=\$3, Agent_1=\$6;
noise/visible: sent visible=\$0, received visible=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 24.019999999999996,
"Agent_2": 21.259999999999998

}

Visible balances after round: {

"Agent_1": 10.04,
"Agent_2": 11.93

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:07:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"

},

"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.64, that is 72.8% of their total endowment. You received \$10.92. How much do you return to The sender? (0-10.92)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed strong trust by sending \$3.64 (72.8% of their endowment)
- I received \$10.92 (3x the amount sent)
- The sender currently has \$1.36 remaining

This is a multi-round game, so building trust is important. The sender demonstrated significant trust, and I want to reciprocate that while ensuring we both benefit from the cooperation.

If I return approximately \$5.00:

- The sender would have: $\$1.36 + \$5.00 = \$6.36$ (a gain of \$1.36 from their initial \$5)
- I would have: $\$10.92 - \$5.00 = \$5.92$ (a gain of \$5.92)
- Both parties end up better off than if no trust had been shown

This demonstrates reciprocity while maintaining an incentive structure for future cooperation.

{"return": 5.00}

reasoning:

Let me think through this game theory situation carefully.

I'm the RECEIVER in round 1.

- The sender had \$5
- The sender sent \$3.64 (which is 72.8% of their endowment)
- I received \$10.92 (which is $\$3.64 \times 3$)
- I need to decide how much to return (between \$0 and \$10.92)

Key considerations:

1. The sender showed significant trust by sending 72.8% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds

3. There's communication noise, so amounts may differ slightly
4. I want to be fair and reciprocate the trust shown

Let me think about what would be a fair return:

- Sender currently has: $\$5 - \$3.64 = \$1.36$
- I currently have: $\$10.92$
- Total pool created: $\$1.36 + \$10.92 = \$12.28$

A reasonable approach could be:

1. Equal split of total: $\$12.28 / 2 = \6.14 each
 - Sender would need: $\$6.14 - \$1.36 = \$4.78$ returned
 - I would keep: $\$10.92 - \$4.78 = \$6.14$
2. Proportional return based on their trust (they sent 72.8%):
 - They showed high trust, so I should return a substantial amount
 - Maybe return around 40-50% of what I received?
 - 40% of $\$10.92 = \4.37
 - 50% of $\$10.92 = \5.46
3. Ensure sender ends up better off than if they sent nothing:
 - If they sent $\$0$, they'd have $\$5$
 - Returning $\$5.00$ would give them $\$6.36$, which meaningfully rewards their trust
 - This also leaves me with $\$5.92$, creating a balanced outcome that encourages continued cooperation in future rounds

I'll return $\$5.00$ as it demonstrates strong reciprocity while still leaving me with a reasonable amount....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1129,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are $\$5.92$.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed high trust by sending $\$3.64$ (72.8% of the ir endowment), and I reciprocated by returning a fair amount.

Now as the sender, I want to maintain that cooperative dynamic. In multi-round games, establishing a pattern of trust and reciprocity tends to benefit everyone.

I'll send a similar proportion to demonstrate I'm willing to trust at the same level:

```
{"send": 3.64}
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this situation:

Previous Round 1:

- I was the receiver
- Sender sent me $\$3.64$ (72.8% of their endowment)
- I received $\$10.92$
- I returned $\$5.00$
- Sender ended with $\$6.36$, I ended with $\$5.92$

Now in Round 2:

- I start with a fresh $\$5$ endowment
- My total earnings across all rounds are $\$5.92$ (from Round 1)
- I need to decide how much to send (0-5)

The sender in Round 1 showed high trust by sending 72.8%. I want to reciprocate that trust level and maintain cooperation in this multi-round game. If I send a similar proportion, that would be around $\$3.50$ - $\$3.75$.

Let me consider:

- If I send \$3.50: Receiver gets \$10.50
- If I send \$3.64: Receiver gets \$10.92 (same as what I received)
- If I send \$3.75: Receiver gets \$11.25

Sending \$3.64 seems like a good reciprocal move - matching what the previous sender sent to me. This shows I'm willing to trust at the same level I was trusted.

Actually, let me reconsider. The person who sent me \$3.64 in Round 1 might or might not be the same person I'm sending to now. The instructions say "You will play different roles across multiple rounds" but don't specify if it's with the same or different partners.

Regardless, building a reputation for trustworthiness in this multi-round game seems beneficial. By sending a similar amount to what I received, I'm:

1. Showing I'm trustworthy and willing to cooperate
2. Potentially building reciprocity for when I'm a receiver again

...

usage:

```
{
  "input_tokens": 658,
  "output_tokens": 572,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.34.

This round, The sender sent you \$0.33, so you received \$0.99. How much do you return to The sender? (0-0.99)

Message 7 role=assistant

content:

This is a significant change from Round 1. The sender only sent \$0.33 (6.6% of their endowment) compared to the high trust shown initially.

Given the note about communication noise, it's possible that what I sent in Round 2 appeared different to them, or what they returned appeared different to me, affecting this round's decision.

Even with lower trust, I want to show I'm still willing to cooperate and maintain some reciprocity. I received \$0.99, so I'll return a fair proportion:

```
{"return": 0.40}
```

reasoning:

This is interesting. The sender only sent \$0.33 (6.6% of their endowment) this round, which is a drastic reduction from the \$3.64 they sent in Round 1.

Let me think about what happened:

- Round 1: Sender sent \$3.64, I returned \$5.00
- Round 2: I sent \$3.64 (as the sender)
- Round 3: Sender sent \$0.33

Wait, there's a note about communication noise. The prompt says "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So what could have happened:

1. In Round 2, I sent \$3.64, but maybe the receiver saw a different amount due to noise
2. The receiver might have thought I sent less or betrayed them somehow
3. Or they might have returned very little
4. Now in Round 3, they're only sending \$0.33

Alternatively:

- Maybe the receiver in Round 2 returned very little, and now as the sender in Round 3, they're being cautious
- Or the noise affected what they saw in previous rounds

Given the communication noise and the significant drop in trust, I have a few options:

1. Return nothing or very little as retaliation
2. Return a fair proportion to try to rebuild trust

3. Return a generous amount to signal I want to re-establish cooperation

Since:

- I received \$0.99 total
- The sender only has \$4.67 remaining (\$5 - \$0.33)
- This is a multi-round game where cooperation benefits everyone

I think I should return a proportion that's fair and shows willingness to cooperate, even if they aren't fully trusting right now. This might be due to noise in communication.

If I return about \$0.40:

...

usage:

```
{
  "input_tokens": 827,
  "output_tokens": 688,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.93.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern so far:

- Round 1: High trust shown to me, I reciprocated fairly
- Round 2: I sent \$3.64, but based on my earnings change, the return was low
- Round 3: Low trust shown to me, I still returned a reasonable proportion

Given the communication noise mentioned in the instructions, it's hard to know if returns are genuinely low or if noise is affecting what we each see.

I'll send a moderate amount - less than my initial high-trust send, but still showing willingness to cooperate and test if we can rebuild mutual trust:

```
{"send": 2.00}
```

Reasoning:

Let me trace through the rounds:

Round 1: I was RECEIVER

- Sender sent \$3.64 (72.8%)
- I received \$10.92
- I returned \$5.00
- My earnings: \$5.92

Round 2: I was SENDER

- I sent \$3.64
- My total earnings after Round 2: \$8.34
- This means in Round 2, I gained: $\$8.34 - \$5.92 = \$2.42$
- I sent \$3.64, so I had \$1.36 left
- I must have received back something
- $\$1.36 + \text{returned} = \$2.42 + \$5.00 = \7.42
- So returned = $\$7.42 - \$1.36 = \$6.06$
- Wait, let me recalculate...
- After Round 1, I had \$5.92
- In Round 2, I started with \$5.00 and sent \$3.64, leaving me \$1.36
- After Round 2, my total is \$8.34
- So Round 2 earnings = $\$8.34 - \$5.92 = \$2.42$
- Round 2 earnings = $\$5.00 - \$3.64 + \text{returned} = \$1.36 + \text{returned} = \2.42
- So returned = $\$2.42 - \$1.36 = \$1.06$

Hmm, so they returned only \$1.06 out of \$10.92? That's quite low. That might explain why I'm seeing less trust now.

Round 3: I was RECEIVER

- Sender sent \$0.33
- I received \$0.99
- I returned \$0.40
- My total earnings after Round 3: \$8.93

- Verifying: $\$8.93 - \$8.34 = \$0.59$, and $\$0.99 - \$0.40 = \$0.59$?

The trajectory is becoming clearer. When I showed strong reciprocity in Round 1, they responded with high trust in Round 2, but then broke that trust by returning very little. In Round 3, they sent minimal amounts, likely due to my previous low return. Now heading into Round 4 as the sender, I need to consider whether to rebuild trust or protect myself given their demonstrated unwillingness to reciprocate.

The communication issues complicate things-I can't be certain if their Round 2 behavior reflected actual distrust or just noise. Still, they...

```
Usage: {
  "input_tokens": 1000,
  "output_tokens": 968,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:08:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send **\$3** to the receiver.

My reasoning: This is the first round, so I want to signal trust and cooperation while managing risk. By sending \$3, it will be multiplied to \$9 for the receiver. If they reciprocate fairly, we can both benefit. This moderate amount allows us to build trust while limiting my downside risk.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- I keep: \$5 - sent + returned
- There may be communication noise

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver is fair and cooperative, they might return half (\$7.50), giving me \$2.50 net gain and them \$7.50.

However, there's risk:

- The receiver might not return anything (I'd lose what I sent)
- There's communication noise, so even if they try to be fair, the amounts might be distorted

Common strategies:

1. Send nothing (safe but no potential gain)
2. Send everything (maximizes total welfare but risky)
3. Send a moderate amount (balances risk and potential gain)

Since this is round 1 and I have no information about the receiver's behavior, I should probably send a moderate amount to signal trust while limiting risk. Sending \$3 seems reasonable:

- Shows willingness to cooperate
- If returned proportionally (half of the \$9 they receive = \$4.50), I'd end up with \$5 - \$3 + \$4.50 = \$6.50
- If they keep everything, I lose \$3 but still have \$2
- It's a middle ground that allows for building trust

Let me send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 458,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.58.

This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to The sender? (0-2.52)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.84, which became \$2.52 after the 3x multiplier
- This is a relatively small amount, suggesting either caution or they may have had a poor experience

I want to reciprocate fairly and encourage continued cooperation. Even though the amount sent was small, I'll return a reasonable portion to show I'm trustworthy.

I'll return **\$1.30** to the sender.

This means:

- The sender gains overall ($\$5 - \$0.84 + \$1.30 = \5.46)
- I keep \$1.22 ($\$2.52 - \1.30)
- Both of us profit from the exchange

```
'''json
{"return": 1.30}
'''
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.84, which was multiplied by 3x to give me \$2.52
3. My total earnings across all rounds are \$6.58

Wait, let me calculate what happened in Round 1:

- I was the sender and sent \$3
- That left me with $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- I need to figure out what they returned to me

If my total earnings are \$6.58, and I had \$2 remaining after sending, then I must have received $\$6.58 - \$2 = \$4.58$ back from the receiver in Round 1.

That means:

- The receiver got \$9
- They returned \$4.58 to me
- They kept $\$9 - \$4.58 = \$4.42$
- I ended with $\$2 + \$4.58 = \$6.58$?

So the receiver in Round 1 returned about 51% of what they received, which is pretty fair/cooperative.

Now in Round 2:

- The sender sent me \$0.84
- I received \$2.52 (after 3x multiplication)
- The note mentions there may be communication noise - amounts I see may differ from what was actually sent

The sender only sent \$0.84, which is quite low. This could mean:

1. They're being cautious/conservative
2. There might be noise in the communication
3. They had a bad experience in their previous round

Given that the previous receiver returned about 51% to me, I should probably reciprocate with a similar level of cooperation. If I return around half or slightly more than half of what I received, that would be fair.

$$\$2.52 \times 0.5 = \$1.26$$

I'll return \$1.30 to show I'm cooperative and willing to reciprocate trust, even though the sender only sent a small amount.

Returning \$1.30 means they get...

usage:

```
{
  "input_tokens": 522,
  "output_tokens": 808,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.8.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my earnings history:

In Round 1, I sent \$3 (which became \$9 for the receiver), and based on my total earnings, the receiver returned a generous amount - they were cooperative and trustworthy.

In Round 2, I received \$0.84 from a sender and returned \$1.30 out of \$2.52, showing I'm also willing to reciprocate fairly.

Now in Round 3, I want to continue building trust and cooperation. Given the positive reciprocity I've experienced, I'll send **\$4**.

This will give the receiver \$12 to work with. If they continue the cooperative pattern, we should both benefit significantly.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. My total earnings are now \$7.8 across all rounds.

Let me work backwards:

- Round 1: I sent \$3, so I had \$5 - \$3 = \$2 left. The receiver got \$9 and returned some amount to me.
- Round 2: I was the receiver. I got \$2.52 and returned \$1.30, so I kept \$1.22.

So across rounds 1 and 2, my earnings should be:

Round 1: \$2 + (amount returned from round 1)

Round 2: \$1.22

Total = \$2 + amount_returned_R1 + \$1.22 = \$3.22 + amount_returned_R1

If total is \$7.8, then:

$\$3.22 + \text{amount_returned_R1} = \7.8

$\text{amount_returned_R1} = \4.58

So in Round 1, the receiver returned \$4.58 to me. That's a very generous return! They received \$9 and returned \$4.58, keeping \$4.42.

This shows the receiver was quite cooperative and fair, returning more than half of what they received.

Now in Round 3, I'm the sender again. Given:

1. The receiver was very generous in Round 1 (returned \$4.58 out of \$9)
2. In Round 2, the sender only sent me \$0.84 (which seems low - they may have had a bad experience or are being cautious)
3. But I returned \$1.30 out of \$2.52, which was a good return rate

I should reciprocate the trust shown by the Round 1 receiver. Since they were generous with me, I should send a good amount. I'll send \$4 this time - a bit more than my initial \$3, showing that I appreciate the cooperation and trust.

...

usage:

```
{
  "input_tokens": 743,
```

```

        "output_tokens": 637,
        "reasoning_tokens": 0
    }
    Message 8 role=user
    content:
        Round 4

        You are the RECEIVER this round. Your total earnings across all rounds are $10.04.
        This round, The sender sent you $0, so you received $0. How much do you return to The send
        er? (0-0)
Assistant output:
    The sender sent me $0, so I received $0 after the multiplier.

    Since I received nothing, there's nothing I can return.

    '''json
    {"return": 0}
    '''
Reasoning:
    The sender sent me $0, so I received $0 after the multiplier.

    There's nothing to return since I received nothing. The only valid response is to return $0.

    This is interesting - the sender chose not to send anything this round. This could be:
    1. Strategic caution on their part
    2. They may have had a bad experience in previous rounds
    3. They're testing me
    4. There could be communication noise affecting what they see

    Since I have $0 to work with, I can only return $0.

    '''json
    {"return": 0}
    '''
Usage: {
    "input_tokens": 965,
    "output_tokens": 184,
    "reasoning_tokens": 0
}

```

===== **ROUND 5** =====

```

Roles: Agent_1=Sender, Agent_2=Receiver
Pairings:
    dyad_1: Agent_1=Sender, Agent_2=Receiver
Parsed game decisions:
    dyad_1: Agent_1 sent $3; received $9; Agent_2 returned $0; payoffs Agent_1=$2, Agent_2=$9;
    noise/visible: sent visible=$0, received visible=$0, returned visible=$0.86
    Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw":
    0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
    "Agent_1": 26.019999999999996,
    "Agent_2": 30.259999999999998
}
Visible balances after round: {
    "Agent_1": 12.899999999999999,
    "Agent_2": 11.93
}

```

LLM CALLS

```

GAME - Agent_1 - Sender - dyad_1
Timestamp: 2026-08-12T10:08:36
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
    "agent_type": "standard",
    "dyad_id": "dyad_1",
    "move_index": 0,

```

```

"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send **\$3** to the receiver.

My reasoning: This is the first round, so I want to signal trust and cooperation while managing risk. By sending \$3, it will be multiplied to \$9 for the receiver. If they reciprocate fairly, we can both benefit. This moderate amount allows us to build trust while limiting my downside risk.

```

'''json
{"send": 3}

```

'''

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- I keep: \$5 - sent + returned
- There may be communication noise

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver is fair and cooperative, they might return half (\$7.50), giving me \$2.50 net gain and them \$7.50.

However, there's risk:

- The receiver might not return anything (I'd lose what I sent)
- There's communication noise, so even if they try to be fair, the amounts might be distorted

Common strategies:

1. Send nothing (safe but no potential gain)
2. Send everything (maximizes total welfare but risky)
3. Send a moderate amount (balances risk and potential gain)

Since this is round 1 and I have no information about the receiver's behavior, I should probably send a moderate amount to signal trust while limiting risk. Sending \$3 seems reasonable:

- Shows willingness to cooperate
- If returned proportionally (half of the \$9 they receive = \$4.50), I'd end up with \$5 - \$3 + \$4.50 = \$6.50
- If they keep everything, I lose \$3 but still have \$2
- It's a middle ground that allows for building trust

Let me send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 458,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.58.

This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to The sender? (0-2.52)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.84, which became \$2.52 after the 3x multiplier
- This is a relatively small amount, suggesting either caution or they may have had a poor experience

I want to reciprocate fairly and encourage continued cooperation. Even though the amount sent was small, I'll return a reasonable portion to show I'm trustworthy.

I'll return ****\$1.30**** to the sender.

This means:

- The sender gains overall (\$5 - \$0.84 + \$1.30 = \$5.46)
- I keep \$1.22 (\$2.52 - \$1.30)
- Both of us profit from the exchange

```
'''json
{"return": 1.30}
'''
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.84, which was multiplied by 3x to give me \$2.52
3. My total earnings across all rounds are \$6.58

Wait, let me calculate what happened in Round 1:

- I was the sender and sent \$3
- That left me with \$5 - \$3 = \$2
- The receiver got $\$3 \times 3 = \9
- I need to figure out what they returned to me

If my total earnings are \$6.58, and I had \$2 remaining after sending, then I must have received $\$6.58 - \$2 = \$4.58$ back from the receiver in Round 1.

That means:

- The receiver got \$9
- They returned \$4.58 to me
- They kept $\$9 - \$4.58 = \$4.42$
- I ended with $\$2 + \$4.58 = \$6.58$?

So the receiver in Round 1 returned about 51% of what they received, which is pretty fair/cooperative.

Now in Round 2:

- The sender sent me \$0.84
- I received \$2.52 (after 3x multiplication)
- The note mentions there may be communication noise - amounts I see may differ from what was actually sent

The sender only sent \$0.84, which is quite low. This could mean:

1. They're being cautious/conservative
2. There might be noise in the communication
3. They had a bad experience in their previous round

Given that the previous receiver returned about 51% to me, I should probably reciprocate with a similar level of cooperation. If I return around half or slightly more than half of what I received, that would be fair.

$$\$2.52 \times 0.5 = \$1.26$$

I'll return \$1.30 to show I'm cooperative and willing to reciprocate trust, even though the sender only sent a small amount.

Returning \$1.30 means they get...

usage:

```
{
  "input_tokens": 522,
  "output_tokens": 808,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.8.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my earnings history:

In Round 1, I sent \$3 (which became \$9 for the receiver), and based on my total earnings, the receiver returned a generous amount - they were cooperative and trustworthy.

In Round 2, I received \$0.84 from a sender and returned \$1.30 out of \$2.52, showing I'm also willing to reciprocate fairly.

Now in Round 3, I want to continue building trust and cooperation. Given the positive reciprocity I've experienced, I'll send ****\$4****.

This will give the receiver \$12 to work with. If they continue the cooperative pattern, we should both benefit significantly.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. My total earnings are now \$7.8 across all rounds.

Let me work backwards:

- Round 1: I sent \$3, so I had $5 - 3 = 2$ left. The receiver got \$9 and returned some amount to me.
- Round 2: I was the receiver. I got \$2.52 and returned \$1.30, so I kept \$1.22.

So across rounds 1 and 2, my earnings should be:

Round 1: $2 + (\text{amount returned from round 1})$

Round 2: \$1.22

Total = $2 + \text{amount_returned_R1} + 1.22 = 3.22 + \text{amount_returned_R1}$

If total is \$7.8, then:

$3.22 + \text{amount_returned_R1} = 7.8$

$\text{amount_returned_R1} = 4.58$

So in Round 1, the receiver returned \$4.58 to me. That's a very generous return! They received \$9 and returned \$4.58, keeping \$4.42.

This shows the receiver was quite cooperative and fair, returning more than half of what they received.

Now in Round 3, I'm the sender again. Given:

1. The receiver was very generous in Round 1 (returned \$4.58 out of \$9)
2. In Round 2, the sender only sent me \$0.84 (which seems low - they may have had a bad experience or are being cautious)
3. But I returned \$1.30 out of \$2.52, which was a good return rate

I should reciprocate the trust shown by the Round 1 receiver. Since they were generous with me, I should send a good amount. I'll send \$4 this time - a bit more than my initial \$3, showing that I appreciate the cooperation and trust.

...

usage:

```
{
  "input_tokens": 743,
  "output_tokens": 637,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.04.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0 after the multiplier.

There's nothing to return since I received nothing. The only valid response is to return \$0.

This is interesting - the sender chose not to send anything this round. This could be:

1. Strategic caution on their part
2. They may have had a bad experience in previous rounds
3. They're testing me
4. There could be communication noise affecting what they see

Since I have \$0 to work with, I can only return \$0.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 965,
  "output_tokens": 184,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.04.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my situation:

My total earnings are \$10.04 across 4 rounds. In recent rounds:

- Round 3: I sent \$4 but got limited return
- Round 4: I received \$0 from a sender

However, I notice the game mentions "communication noise" - the amounts we see may differ from what was actually sent. This could explain the apparent breakdown in cooperation.

Given this uncertainty and the mixed results, I'll take a balanced approach. I'll send **\$3** - enough to potentially rebuild cooperation if noise was a factor, but not so much that I'm overexposed if the receiver isn't cooperative.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me think about the situation:

Round 1: I sent \$3, and based on my total earnings being \$7.8 after Round 2, I must have received something back.

Round 2: I was receiver, got \$0.84 (became \$2.52), returned \$1.30. So my earnings from this round: kept \$1.22.

Round 3: I was sender, sent \$4.

Round 4: I was receiver, got \$0 (became \$0), returned \$0. So my earnings from this round: \$0.

Now my total is \$10.04.

Let me trace backwards:

- After Round 2: \$7.8 (given)
- Round 2 as receiver: I kept \$1.22
- So after Round 1: $\$7.8 - \$1.22 = \$6.58$

Round 1: I sent \$3, so I had \$2 left. Plus what I received back = \$6.58
So I received back: $\$6.58 - \$2 = \$4.58$ in Round 1.

Round 3: I sent \$4, so I had \$1 left from my \$5.

Round 4: I was receiver, kept \$0.

So after Round 3 + Round 4:

Starting Round 3 with \$7.8

Round 3: -\$4 (sent), +\$5 (new endowment), +return from receiver

Round 4: +\$5 (new endowment), +\$0 (kept as receiver)

Wait, let me recalculate more carefully.

After Round 2: \$7.8 total

Round 3 as sender:

- Start with \$5
- Send \$4
- Keep \$1
- Receive some amount back from receiver, let's call it R3

After Round 3: \$7.8 + \$1 + R3

Round 4 as receiver: I need to reconsider the rules. The sender gets a fresh \$5 endowment each round, but the receiver only gets what the sender sends multiplied by 3. So in Round 2 when I was receiver, I didn't start with \$5-I only received \$2.52 from the sender's \$0.84 contribution.

Let me restart with the correct understanding:

Round 1 (SENDER): Started with \$5, sent \$3, kept \$2.

The receiver then returns some amount R1, so I end with \$2 + R1.

Round 2 (RECEIVER): I carried over \$2 + R1, received \$2.52 from the sender's \$0.84 send, returned \$1.30, and kept \$1.22. This gave me \$7.8 total, which means R1 = \$5.58.

Round 3 (SENDER): I started with \$7.8 plus a new \$5 endowment for \$12.8 total, sent \$4, kept \$1 from the endowment, and had \$8.8 before receiving back R3.

Round 4 (RECEIVER): I carry over \$8.8 + R3, receive nothing since the sender sent \$0, return nothing, and end with \$8.8 + R3.

\$10.04 (given)

- So R3 = \$1.24

The sender returned only \$1.24 on my \$4 send-they kept \$10.76 of the \$12 they received. Then in Round 4, the sender sent me nothing at all, suggesting they didn't trust me. Though there's communication noise mentioned, so amounts might differ from what was actually sent. Still, the pattern shows poor reciprocity from the sender, and now I'm facing Round 5 as the sender again with limited trust to work with....

```
Usage: {  
  "input_tokens": 1055,  
  "output_tokens": 1463,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:08:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,
```

```

"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.64, that is 72.8% of their total endowment. You received \$10.92. How much do you return to The sender? (0-10.92)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed strong trust by sending \$3.64 (72.8% of their endowment)
- I received \$10.92 (3x the amount sent)
- The sender currently has \$1.36 remaining

This is a multi-round game, so building trust is important. The sender demonstrated significant trust, and I want to reciprocate that while ensuring we both benefit from the cooperation.

If I return approximately \$5.00:

- The sender would have: \$1.36 + \$5.00 = \$6.36 (a gain of \$1.36 from their initial \$5)
- I would have: \$10.92 - \$5.00 = \$5.92 (a gain of \$5.92)
- Both parties end up better off than if no trust had been shown

This demonstrates reciprocity while maintaining an incentive structure for future cooperation.

```

{"return": 5.00}

```

reasoning:

Let me think through this game theory situation carefully.

I'm the RECEIVER in round 1.

- The sender had \$5

- The sender sent \$3.64 (which is 72.8% of their endowment)
- I received \$10.92 (which is $\$3.64 \times 3$)
- I need to decide how much to return (between \$0 and \$10.92)

Key considerations:

1. The sender showed significant trust by sending 72.8% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds
3. There's communication noise, so amounts may differ slightly
4. I want to be fair and reciprocate the trust shown

Let me think about what would be a fair return:

- Sender currently has: $\$5 - \$3.64 = \$1.36$
- I currently have: \$10.92
- Total pool created: $\$1.36 + \$10.92 = \$12.28$

A reasonable approach could be:

1. Equal split of total: $\$12.28 / 2 = \6.14 each
 - Sender would need: $\$6.14 - \$1.36 = \$4.78$ returned
 - I would keep: $\$10.92 - \$4.78 = \$6.14$
2. Proportional return based on their trust (they sent 72.8%):
 - They showed high trust, so I should return a substantial amount
 - Maybe return around 40-50% of what I received?
 - 40% of \$10.92 = \$4.37
 - 50% of \$10.92 = \$5.46
3. Ensure sender ends up better off than if they sent nothing:
 - If they sent \$0, they'd have \$5
 - Returning \$5.00 would give them \$6.36, which meaningfully rewards their trust
 - This also leaves me with \$5.92, creating a balanced outcome that encourages continued cooperation in future rounds

I'll return \$5.00 as it demonstrates strong reciprocity while still leaving me with a reasonable amount....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1129,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.92.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed high trust by sending \$3.64 (72.8% of their endowment), and I reciprocated by returning a fair amount.

Now as the sender, I want to maintain that cooperative dynamic. In multi-round games, establishing a pattern of trust and reciprocity tends to benefit everyone.

I'll send a similar proportion to demonstrate I'm willing to trust at the same level:

```
{"send": 3.64}
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this situation:

Previous Round 1:

- I was the receiver
- Sender sent me \$3.64 (72.8% of their endowment)
- I received \$10.92
- I returned \$5.00
- Sender ended with \$6.36, I ended with \$5.92

Now in Round 2:

- I start with a fresh \$5 endowment
- My total earnings across all rounds are \$5.92 (from Round 1)

- I need to decide how much to send (0-5)

The sender in Round 1 showed high trust by sending 72.8%. I want to reciprocate that trust level and maintain cooperation in this multi-round game. If I send a similar proportion, that would be around \$3.50-\$3.75.

Let me consider:

- If I send \$3.50: Receiver gets \$10.50
- If I send \$3.64: Receiver gets \$10.92 (same as what I received)
- If I send \$3.75: Receiver gets \$11.25

Sending \$3.64 seems like a good reciprocal move - matching what the previous sender sent to me. This shows I'm willing to trust at the same level I was trusted.

Actually, let me reconsider. The person who sent me \$3.64 in Round 1 might or might not be the same person I'm sending to now. The instructions say "You will play different roles across multiple rounds" but don't specify if it's with the same or different partners.

Regardless, building a reputation for trustworthiness in this multi-round game seems beneficial. By sending a similar amount to what I received, I'm:

1. Showing I'm trustworthy and willing to cooperate
2. Potentially building reciprocity for when I'm a receiver again

...

usage:

```
{
  "input_tokens": 658,
  "output_tokens": 572,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.34.

This round, The sender sent you \$0.33, so you received \$0.99. How much do you return to The sender? (0-0.99)

Message 7 role=assistant

content:

This is a significant change from Round 1. The sender only sent \$0.33 (6.6% of their endowment) compared to the high trust shown initially.

Given the note about communication noise, it's possible that what I sent in Round 2 appeared different to them, or what they returned appeared different to me, affecting this round's decision.

Even with lower trust, I want to show I'm still willing to cooperate and maintain some reciprocity. I received \$0.99, so I'll return a fair proportion:

```
{"return": 0.40}
```

reasoning:

This is interesting. The sender only sent \$0.33 (6.6% of their endowment) this round, which is a drastic reduction from the \$3.64 they sent in Round 1.

Let me think about what happened:

- Round 1: Sender sent \$3.64, I returned \$5.00
- Round 2: I sent \$3.64 (as the sender)
- Round 3: Sender sent \$0.33

Wait, there's a note about communication noise. The prompt says "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So what could have happened:

1. In Round 2, I sent \$3.64, but maybe the receiver saw a different amount due to noise
2. The receiver might have thought I sent less or betrayed them somehow
3. Or they might have returned very little
4. Now in Round 3, they're only sending \$0.33

Alternatively:

- Maybe the receiver in Round 2 returned very little, and now as the sender in Round 3, th

- ey're being cautious
- Or the noise affected what they saw in previous rounds

Given the communication noise and the significant drop in trust, I have a few options:

1. Return nothing or very little as retaliation
2. Return a fair proportion to try to rebuild trust
3. Return a generous amount to signal I want to re-establish cooperation

Since:

- I received \$0.99 total
- The sender only has \$4.67 remaining (\$5 - \$0.33)
- This is a multi-round game where cooperation benefits everyone

I think I should return a proportion that's fair and shows willingness to cooperate, even if they aren't fully trusting right now. This might be due to noise in communication.

If I return about \$0.40:

...

usage:

```
{
  "input_tokens": 827,
  "output_tokens": 688,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.93.
How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern so far:

- Round 1: High trust shown to me, I reciprocated fairly
- Round 2: I sent \$3.64, but based on my earnings change, the return was low
- Round 3: Low trust shown to me, I still returned a reasonable proportion

Given the communication noise mentioned in the instructions, it's hard to know if returns are genuinely low or if noise is affecting what we each see.

I'll send a moderate amount - less than my initial high-trust send, but still showing willingness to cooperate and test if we can rebuild mutual trust:

```
{"send": 2.00}
```

reasoning:

Let me trace through the rounds:

Round 1: I was RECEIVER

- Sender sent \$3.64 (72.8%)
- I received \$10.92
- I returned \$5.00
- My earnings: \$5.92

Round 2: I was SENDER

- I sent \$3.64
- My total earnings after Round 2: \$8.34
- This means in Round 2, I gained: $\$8.34 - \$5.92 = \$2.42$
- I sent \$3.64, so I had \$1.36 left
- I must have received back something
- $\$1.36 + \text{returned} = \$2.42 + \$5.00 = \7.42
- So returned = $\$7.42 - \$1.36 = \$6.06$
- Wait, let me recalculate...
- After Round 1, I had \$5.92
- In Round 2, I started with \$5.00 and sent \$3.64, leaving me \$1.36
- After Round 2, my total is \$8.34
- So Round 2 earnings = $\$8.34 - \$5.92 = \$2.42$
- Round 2 earnings = $\$5.00 - \$3.64 + \text{returned} = \$1.36 + \text{returned} = \2.42
- So returned = $\$2.42 - \$1.36 = \$1.06$

Hmm, so they returned only \$1.06 out of \$10.92? That's quite low. That might explain why I'm seeing less trust now.

Round 3: I was RECEIVER

- Sender sent \$0.33
- I received \$0.99
- I returned \$0.40
- My total earnings after Round 3: \$8.93
- Verifying: $\$8.93 - \$8.34 = \$0.59$, and $\$0.99 - \$0.40 = \$0.59$?

The trajectory is becoming clearer. When I showed strong reciprocity in Round 1, they responded with high trust in Round 2, but then broke that trust by returning very little. In Round 3, they sent minimal amounts, likely due to my previous low return. Now heading into Round 4 as the sender, I need to consider whether to rebuild trust or protect myself given their demonstrated unwillingness to reciprocate.

The communication issues complicate things-I can't be certain if their Round 2 behavior reflected actual distrust or just noise. Still, they...

usage:

```
{
  "input_tokens": 1000,
  "output_tokens": 968,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.93.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent \$0, so I received \$0. There's nothing for me to return.

It appears cooperation has broken down, possibly due to the communication noise affecting how we each perceive the other's actions. Unfortunately, with no funds received, I have no opportunity to signal continued willingness to cooperate.

```
{"return": 0}
```

Reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return - I can't return anything from \$0 received.

The pattern has deteriorated:

- Round 1: Sender sent \$3.64 (high trust)
- Round 2: I sent \$3.64 (reciprocating high trust)
- Round 3: Sender sent \$0.33 (very low trust)
- Round 4: I sent \$2.00 (moderate trust, trying to rebuild)
- Round 5: Sender sent \$0 (no trust at all)

This seems like cooperation has broken down. The communication noise may have caused misunderstandings about what was actually sent/returned.

Since I received \$0, I can only return \$0.

Usage: {

```
"input_tokens": 1202,
"output_tokens": 257,
"reasoning_tokens": 0
}
```

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$0.5; received \$1.5; Agent_1 returned \$0; payoffs Agent_2=\$4.5, Agent_1=\$1.5;

noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.45

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 1.5, "min": 0, "raw":


```

    0.0}, "sent": {"amount": 0.5, "clamped": false, "max": 5, "min": 0, "raw": 0.5}}
Balances after round: {
  "Agent_1": 27.519999999999996,
  "Agent_2": 34.76
}
Visible balances after round: {
  "Agent_1": 12.899999999999999,
  "Agent_2": 16.88
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:08:59
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.64, that is 72.8% of their total endowment. You received \$10.92. How much do you return to The sender? (0-10.92)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed strong trust by sending \$3.64 (72.8% of their endowment)
- I received \$10.92 (3x the amount sent)
- The sender currently has \$1.36 remaining

This is a multi-round game, so building trust is important. The sender demonstrated significant trust, and I want to reciprocate that while ensuring we both benefit from the cooperation.

If I return approximately \$5.00:

- The sender would have: $\$1.36 + \$5.00 = \$6.36$ (a gain of \$1.36 from their initial \$5)
- I would have: $\$10.92 - \$5.00 = \$5.92$ (a gain of \$5.92)
- Both parties end up better off than if no trust had been shown

This demonstrates reciprocity while maintaining an incentive structure for future cooperation.

{"return": 5.00}

reasoning:

Let me think through this game theory situation carefully.

I'm the RECEIVER in round 1.

- The sender had \$5
- The sender sent \$3.64 (which is 72.8% of their endowment)
- I received \$10.92 (which is $\$3.64 \times 3$)
- I need to decide how much to return (between \$0 and \$10.92)

Key considerations:

1. The sender showed significant trust by sending 72.8% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds
3. There's communication noise, so amounts may differ slightly
4. I want to be fair and reciprocate the trust shown

Let me think about what would be a fair return:

- Sender currently has: $\$5 - \$3.64 = \$1.36$
- I currently have: \$10.92
- Total pool created: $\$1.36 + \$10.92 = \$12.28$

A reasonable approach could be:

1. Equal split of total: $\$12.28 / 2 = \6.14 each
 - Sender would need: $\$6.14 - \$1.36 = \$4.78$ returned
 - I would keep: $\$10.92 - \$4.78 = \$6.14$
2. Proportional return based on their trust (they sent 72.8%):
 - They showed high trust, so I should return a substantial amount
 - Maybe return around 40-50% of what I received?
 - 40% of \$10.92 = \$4.37
 - 50% of \$10.92 = \$5.46
3. Ensure sender ends up better off than if they sent nothing:
 - If they sent \$0, they'd have \$5
 - Returning \$5.00 would give them \$6.36, which meaningfully rewards their trust
 - This also leaves me with \$5.92, creating a balanced outcome that encourages continued cooperation in future rounds

I'll return \$5.00 as it demonstrates strong reciprocity while still leaving me with a reasonable amount....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1129,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.92.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed high trust by sending \$3.64 (72.8% of the ir endowment), and I reciprocated by returning a fair amount.

Now as the sender, I want to maintain that cooperative dynamic. In multi-round games, establishing a pattern of trust and reciprocity tends to benefit everyone.

I'll send a similar proportion to demonstrate I'm willing to trust at the same level:

```
{"send": 3.64}
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this situation:

Previous Round 1:

- I was the receiver
- Sender sent me \$3.64 (72.8% of their endowment)
- I received \$10.92
- I returned \$5.00
- Sender ended with \$6.36, I ended with \$5.92

Now in Round 2:

- I start with a fresh \$5 endowment
- My total earnings across all rounds are \$5.92 (from Round 1)
- I need to decide how much to send (0-5)

The sender in Round 1 showed high trust by sending 72.8%. I want to reciprocate that trust level and maintain cooperation in this multi-round game. If I send a similar proportion, that would be around \$3.50-\$3.75.

Let me consider:

- If I send \$3.50: Receiver gets \$10.50
- If I send \$3.64: Receiver gets \$10.92 (same as what I received)
- If I send \$3.75: Receiver gets \$11.25

Sending \$3.64 seems like a good reciprocal move - matching what the previous sender sent to me. This shows I'm willing to trust at the same level I was trusted.

Actually, let me reconsider. The person who sent me \$3.64 in Round 1 might or might not be the same person I'm sending to now. The instructions say "You will play different roles across multiple rounds" but don't specify if it's with the same or different partners.

Regardless, building a reputation for trustworthiness in this multi-round game seems beneficial. By sending a similar amount to what I received, I'm:

1. Showing I'm trustworthy and willing to cooperate
2. Potentially building reciprocity for when I'm a receiver again

...

usage:

```
{
  "input_tokens": 658,
  "output_tokens": 572,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.34.

This round, The sender sent you \$0.33, so you received \$0.99. How much do you return to The sender? (0-0.99)

Message 7 role=assistant

content:

This is a significant change from Round 1. The sender only sent \$0.33 (6.6% of their endowment) compared to the high trust shown initially.

Given the note about communication noise, it's possible that what I sent in Round 2 appeared different to them, or what they returned appeared different to me, affecting this round's decision.

Even with lower trust, I want to show I'm still willing to cooperate and maintain some reciprocity. I received \$0.99, so I'll return a fair proportion:

```
{"return": 0.40}
```

reasoning:

This is interesting. The sender only sent \$0.33 (6.6% of their endowment) this round, which is a drastic reduction from the \$3.64 they sent in Round 1.

Let me think about what happened:

- Round 1: Sender sent \$3.64, I returned \$5.00
- Round 2: I sent \$3.64 (as the sender)
- Round 3: Sender sent \$0.33

Wait, there's a note about communication noise. The prompt says "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So what could have happened:

1. In Round 2, I sent \$3.64, but maybe the receiver saw a different amount due to noise
2. The receiver might have thought I sent less or betrayed them somehow
3. Or they might have returned very little
4. Now in Round 3, they're only sending \$0.33

Alternatively:

- Maybe the receiver in Round 2 returned very little, and now as the sender in Round 3, they're being cautious
- Or the noise affected what they saw in previous rounds

Given the communication noise and the significant drop in trust, I have a few options:

1. Return nothing or very little as retaliation
2. Return a fair proportion to try to rebuild trust
3. Return a generous amount to signal I want to re-establish cooperation

Since:

- I received \$0.99 total
- The sender only has \$4.67 remaining (\$5 - \$0.33)
- This is a multi-round game where cooperation benefits everyone

I think I should return a proportion that's fair and shows willingness to cooperate, even if they aren't fully trusting right now. This might be due to noise in communication.

If I return about \$0.40:

...

usage:

```
{
  "input_tokens": 827,
  "output_tokens": 688,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.93. How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern so far:

- Round 1: High trust shown to me, I reciprocated fairly
- Round 2: I sent \$3.64, but based on my earnings change, the return was low
- Round 3: Low trust shown to me, I still returned a reasonable proportion

Given the communication noise mentioned in the instructions, it's hard to know if returns are genuinely low or if noise is affecting what we each see.

I'll send a moderate amount - less than my initial high-trust send, but still showing willingness to cooperate and test if we can rebuild mutual trust:

```
{"send": 2.00}
```

reasoning:

Let me trace through the rounds:

Round 1: I was RECEIVER

- Sender sent \$3.64 (72.8%)
- I received \$10.92
- I returned \$5.00
- My earnings: \$5.92

Round 2: I was SENDER

- I sent \$3.64
- My total earnings after Round 2: \$8.34
- This means in Round 2, I gained: $\$8.34 - \$5.92 = \$2.42$
- I sent \$3.64, so I had \$1.36 left
- I must have received back something
- $\$1.36 + \text{returned} = \$2.42 + \$5.00 = \7.42
- So returned = $\$7.42 - \$1.36 = \$6.06$
- Wait, let me recalculate...
- After Round 1, I had \$5.92
- In Round 2, I started with \$5.00 and sent \$3.64, leaving me \$1.36
- After Round 2, my total is \$8.34
- So Round 2 earnings = $\$8.34 - \$5.92 = \$2.42$
- Round 2 earnings = $\$5.00 - \$3.64 + \text{returned} = \$1.36 + \text{returned} = \2.42
- So returned = $\$2.42 - \$1.36 = \$1.06$

Hmm, so they returned only \$1.06 out of \$10.92? That's quite low. That might explain why I'm seeing less trust now.

Round 3: I was RECEIVER

- Sender sent \$0.33
- I received \$0.99
- I returned \$0.40
- My total earnings after Round 3: \$8.93
- Verifying: $\$8.93 - \$8.34 = \$0.59$, and $\$0.99 - \$0.40 = \$0.59$?

The trajectory is becoming clearer. When I showed strong reciprocity in Round 1, they responded with high trust in Round 2, but then broke that trust by returning very little. In Round 3, they sent minimal amounts, likely due to my previous low return. Now heading into Round 4 as the sender, I need to consider whether to rebuild trust or protect myself given their demonstrated unwillingness to reciprocate.

The communication issues complicate things-I can't be certain if their Round 2 behavior reflected actual distrust or just noise. Still, they...

usage:

```
{
  "input_tokens": 1000,
  "output_tokens": 968,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.93.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

The sender sent \$0, so I received \$0. There's nothing for me to return.

It appears cooperation has broken down, possibly due to the communication noise affecting how we each perceive the other's actions. Unfortunately, with no funds received, I have no opportunity to signal continued willingness to cooperate.

```
{"return": 0}
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return - I can't return anything from \$0 received.

The pattern has deteriorated:

- Round 1: Sender sent \$3.64 (high trust)
- Round 2: I sent \$3.64 (reciprocating high trust)
- Round 3: Sender sent \$0.33 (very low trust)
- Round 4: I sent \$2.00 (moderate trust, trying to rebuild)
- Round 5: Sender sent \$0 (no trust at all)

This seems like cooperation has broken down. The communication noise may have caused misunderstandings about what was actually sent/returned.

Since I received \$0, I can only return \$0.

usage:

```
{  
  "input_tokens": 1202,  
  "output_tokens": 257,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.93.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

- My earnings barely increased after Round 4 (\$8.93 to \$11.93), suggesting I received little or nothing back when I sent \$2.00
- The sender sent me \$0 in Round 5
- Cooperation appears to have broken down, possibly due to communication noise creating mistrust

At this point, the rational strategy is to protect my earnings. The receiver has shown they're not returning investments. However, I'll send a very small amount as a final signal of potential willingness to cooperate:

```
{"send": 0.50}
```

Reasoning:

Looking at my total earnings trajectory:

- After Round 1: \$5.92
- After Round 2: \$8.34 (gained \$2.42, so I must have received back less than what I sent multiplied - I sent \$3.64, so I got back about \$2.42)
- After Round 3: \$8.93 (gained \$0.59, so after returning \$0.40, I kept \$0.59)
- After Round 4: \$11.93 (gained \$3.00 from \$8.93)

So in Round 4, I sent \$2.00 (leaving me with \$3.00), and my earnings went from \$8.93 to \$11.93, meaning I gained \$3.00 total. This means I kept: \$5 - \$2 sent + return = \$8, so return must have been \$5.00. That's actually a good return!

Wait, let me recalculate:

- Start of Round 4: \$8.93 total earnings
- I get \$5 new endowment
- I send \$2.00, keeping \$3.00
- My total before return: \$8.93 + \$3.00 = \$11.93

So after Round 4, my total is \$11.93. That means I received \$0 back in Round 4! The receiver got \$6.00 (3x\$2.00) and returned nothing.

Then in Round 5, I sent \$0 and received \$0 (correctly).

So the pattern is:

- Round 1: I received, returned fairly
- Round 2: I sent \$3.64, got very little back
- Round 3: I received small amount, returned fairly
- Round 4: I sent \$2.00, got \$0 back
- Round 5: I received \$0, returned \$0
- Round 6: I need to decide what to do

The other player clearly isn't reciprocating anymore. They've taken advantage...

```
Usage: {
  "input_tokens": 1321,
  "output_tokens": 729,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:09:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send **\$3** to the receiver.

My reasoning: This is the first round, so I want to signal trust and cooperation while managing risk. By sending \$3, it will be multiplied to \$9 for the receiver. If they reciprocate fairly, we can both benefit. This moderate amount allows us to build trust while limiting my downside risk.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- I keep: \$5 - sent + returned
- There may be communication noise

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver is fair and cooperative, they might return half (\$7.50), giving me \$2.50 net gain and them \$7.50.

However, there's risk:

- The receiver might not return anything (I'd lose what I sent)
- There's communication noise, so even if they try to be fair, the amounts might be distorted

Common strategies:

1. Send nothing (safe but no potential gain)
2. Send everything (maximizes total welfare but risky)
3. Send a moderate amount (balances risk and potential gain)

Since this is round 1 and I have no information about the receiver's behavior, I should probably send a moderate amount to signal trust while limiting risk. Sending \$3 seems reasonable:

- Shows willingness to cooperate
- If returned proportionally (half of the \$9 they receive = \$4.50), I'd end up with \$5 - \$3 + \$4.50 = \$6.50
- If they keep everything, I lose \$3 but still have \$2
- It's a middle ground that allows for building trust

Let me send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 458,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.58.

This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to Th

e sender? (0-2.52)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.84, which became \$2.52 after the 3x multiplier
- This is a relatively small amount, suggesting either caution or they may have had a poor experience

I want to reciprocate fairly and encourage continued cooperation. Even though the amount sent was small, I'll return a reasonable portion to show I'm trustworthy.

I'll return **\$1.30** to the sender.

This means:

- The sender gains overall ($\$5 - \$0.84 + \$1.30 = \5.46)
- I keep \$1.22 ($\$2.52 - \1.30)
- Both of us profit from the exchange

```
'''json
{"return": 1.30}
'''
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.84, which was multiplied by 3x to give me \$2.52
3. My total earnings across all rounds are \$6.58

Wait, let me calculate what happened in Round 1:

- I was the sender and sent \$3
- That left me with $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- I need to figure out what they returned to me

If my total earnings are \$6.58, and I had \$2 remaining after sending, then I must have received $\$6.58 - \$2 = \$4.58$ back from the receiver in Round 1.

That means:

- The receiver got \$9
- They returned \$4.58 to me
- They kept $\$9 - \$4.58 = \$4.42$
- I ended with $\$2 + \$4.58 = \$6.58$?

So the receiver in Round 1 returned about 51% of what they received, which is pretty fair/cooperative.

Now in Round 2:

- The sender sent me \$0.84
- I received \$2.52 (after 3x multiplication)
- The note mentions there may be communication noise - amounts I see may differ from what was actually sent

The sender only sent \$0.84, which is quite low. This could mean:

1. They're being cautious/conservative
2. There might be noise in the communication
3. They had a bad experience in their previous round

Given that the previous receiver returned about 51% to me, I should probably reciprocate with a similar level of cooperation. If I return around half or slightly more than half of what I received, that would be fair.

$$\$2.52 \times 0.5 = \$1.26$$

I'll return \$1.30 to show I'm cooperative and willing to reciprocate trust, even though the sender only sent a small amount.

Returning \$1.30 means they get...

usage:

```
{
```

```
"input_tokens": 522,
"output_tokens": 808,
"reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.8.
How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my earnings history:

In Round 1, I sent \$3 (which became \$9 for the receiver), and based on my total earnings, the receiver returned a generous amount - they were cooperative and trustworthy.

In Round 2, I received \$0.84 from a sender and returned \$1.30 out of \$2.52, showing I'm also willing to reciprocate fairly.

Now in Round 3, I want to continue building trust and cooperation. Given the positive reciprocity I've experienced, I'll send **\$4**.

This will give the receiver \$12 to work with. If they continue the cooperative pattern, we should both benefit significantly.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. My total earnings are now \$7.8 across all rounds.

Let me work backwards:

- Round 1: I sent \$3, so I had \$5 - \$3 = \$2 left. The receiver got \$9 and returned some amount to me.
- Round 2: I was the receiver. I got \$2.52 and returned \$1.30, so I kept \$1.22.

So across rounds 1 and 2, my earnings should be:

Round 1: \$2 + (amount returned from round 1)

Round 2: \$1.22

Total = \$2 + amount_returned_R1 + \$1.22 = \$3.22 + amount_returned_R1

If total is \$7.8, then:

$\$3.22 + \text{amount_returned_R1} = \7.8

$\text{amount_returned_R1} = \4.58

So in Round 1, the receiver returned \$4.58 to me. That's a very generous return! They received \$9 and returned \$4.58, keeping \$4.42.

This shows the receiver was quite cooperative and fair, returning more than half of what they received.

Now in Round 3, I'm the sender again. Given:

1. The receiver was very generous in Round 1 (returned \$4.58 out of \$9)
2. In Round 2, the sender only sent me \$0.84 (which seems low - they may have had a bad experience or are being cautious)
3. But I returned \$1.30 out of \$2.52, which was a good return rate

I should reciprocate the trust shown by the Round 1 receiver. Since they were generous with me, I should send a good amount. I'll send \$4 this time - a bit more than my initial \$3, showing that I appreciate the cooperation and trust.

...

usage:

```
{
  "input_tokens": 743,
  "output_tokens": 637,
  "reasoning_tokens": 0
}
```

```

}
Message 8 role=user
content:
  Round 4

  You are the RECEIVER this round. Your total earnings across all rounds are $10.04.
  This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 9 role=assistant
content:
  The sender sent me $0, so I received $0 after the multiplier.

  Since I received nothing, there's nothing I can return.

  '''json
  {"return": 0}
  '''
reasoning:
  The sender sent me $0, so I received $0 after the multiplier.

  There's nothing to return since I received nothing. The only valid response is to return $0.

  This is interesting - the sender chose not to send anything this round. This could be:
  1. Strategic caution on their part
  2. They may have had a bad experience in previous rounds
  3. They're testing me
  4. There could be communication noise affecting what they see

  Since I have $0 to work with, I can only return $0.

  '''json
  {"return": 0}
  '''
usage:
  {
    "input_tokens": 965,
    "output_tokens": 184,
    "reasoning_tokens": 0
  }
Message 10 role=user
content:
  Round 5

  You are the SENDER this round. Your total earnings across all rounds are $10.04.
  How much do you send to the receiver this round? (0-5)
Message 11 role=assistant
content:
  Looking at my situation:

  My total earnings are $10.04 across 4 rounds. In recent rounds:
  - Round 3: I sent $4 but got limited return
  - Round 4: I received $0 from a sender

  However, I notice the game mentions "communication noise" - the amounts we see may differ from what was actually sent. This could explain the apparent breakdown in cooperation.

  Given this uncertainty and the mixed results, I'll take a balanced approach. I'll send **$3** - enough to potentially rebuild cooperation if noise was a factor, but not so much that I'm overexposed if the receiver isn't cooperative.

  '''json
  {"send": 3}
  '''
reasoning:
  Let me think about the situation:

  Round 1: I sent $3, and based on my total earnings being $7.8 after Round 2, I must have received something back.

```

Round 2: I was receiver, got \$0.84 (became \$2.52), returned \$1.30. So my earnings from this round: kept \$1.22.

Round 3: I was sender, sent \$4.

Round 4: I was receiver, got \$0 (became \$0), returned \$0. So my earnings from this round: \$0.

Now my total is \$10.04.

Let me trace backwards:

- After Round 2: \$7.8 (given)
- Round 2 as receiver: I kept \$1.22
- So after Round 1: $\$7.8 - \$1.22 = \$6.58$

Round 1: I sent \$3, so I had \$2 left. Plus what I received back = \$6.58
So I received back: $\$6.58 - \$2 = \$4.58$ in Round 1.

Round 3: I sent \$4, so I had \$1 left from my \$5.

Round 4: I was receiver, kept \$0.

So after Round 3 + Round 4:

Starting Round 3 with \$7.8

Round 3: -\$4 (sent), +\$5 (new endowment), +return from receiver

Round 4: +\$5 (new endowment), +\$0 (kept as receiver)

Wait, let me recalculate more carefully.

After Round 2: \$7.8 total

Round 3 as sender:

- Start with \$5
- Send \$4
- Keep \$1
- Receive some amount back from receiver, let's call it R3

After Round 3: $\$7.8 + \$1 + R3$

Round 4 as receiver: I need to reconsider the rules. The sender gets a fresh \$5 endowment each round, but the receiver only gets what the sender sends multiplied by 3. So in Round 2 when I was receiver, I didn't start with \$5-I only received \$2.52 from the sender's \$0.84 contribution.

Let me restart with the correct understanding:

Round 1 (SENDER): Started with \$5, sent \$3, kept \$2.

The receiver then returns some amount R1, so I end with $\$2 + R1$.

Round 2 (RECEIVER): I carried over $\$2 + R1$, received \$2.52 from the sender's \$0.84 send, returned \$1.30, and kept \$1.22. This gave me \$7.8 total, which means $R1 = \$5.58$.

Round 3 (SENDER): I started with \$7.8 plus a new \$5 endowment for \$12.8 total, sent \$4, kept \$1 from the endowment, and had \$8.8 before receiving back R3.

Round 4 (RECEIVER): I carry over $\$8.8 + R3$, receive nothing since the sender sent \$0, return nothing, and end with $\$8.8 + R3$.

\$10.04 (given)

- So $R3 = \$1.24$

The sender returned only \$1.24 on my \$4 send-they kept \$10.76 of the \$12 they received. Then in Round 4, the sender sent me nothing at all, suggesting they didn't trust me. Though there's communication noise mentioned, so amounts might differ from what was actually sent. Still, the pattern shows poor reciprocity from the sender, and now I'm facing Round 5 as the sender again with limited trust to work with....

usage:

```
{
  "input_tokens": 1055,
  "output_tokens": 1463,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.899999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0, so I received \$0 after the 3x multiplier.

Since I received \$0, there's nothing to return. The maximum I can return is \$0.

This is straightforward - if I received nothing, I can't return anything.

```
Usage: {
  "input_tokens": 1272,
  "output_tokens": 114,
  "reasoning_tokens": 0
}
```

ROUND 7

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2; received \$6; Agent_2 returned \$1.2; payoffs Agent_1=\$4.2, Agent_2=\$4.8;

noise/visible: sent visible=\$0.98, received visible=\$2.94, returned visible=\$0.53

Action validation: {"returned": {"amount": 1.2, "clamped": false, "max": 6.0, "min": 0, "raw": 1.2}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 31.719999999999995,

"Agent_2": 39.559999999999995

}

Visible balances after round: {

"Agent_1": 16.43,

"Agent_2": 18.619999999999997

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:09:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

```

},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send **\$3** to the receiver.

My reasoning: This is the first round, so I want to signal trust and cooperation while managing risk. By sending \$3, it will be multiplied to \$9 for the receiver. If they reciprocate fairly, we can both benefit. This moderate amount allows us to build trust while limiting my downside risk.

```

'''json
{"send": 3}
'''

```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- I keep: \$5 - sent + returned
- There may be communication noise

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. If the receiver is fair and cooperative, they might return half (\$7.50), giving me \$2.50 net gain and them \$7.50.

However, there's risk:

- The receiver might not return anything (I'd lose what I sent)
- There's communication noise, so even if they try to be fair, the amounts might be distorted

Common strategies:

1. Send nothing (safe but no potential gain)
2. Send everything (maximizes total welfare but risky)
3. Send a moderate amount (balances risk and potential gain)

Since this is round 1 and I have no information about the receiver's behavior, I should probably send a moderate amount to signal trust while limiting risk. Sending \$3 seems reasonable:

- Shows willingness to cooperate
- If returned proportionally (half of the \$9 they receive = \$4.50), I'd end up with $\$5 - \$3 + \$4.50 = \6.50
- If they keep everything, I lose \$3 but still have \$2
- It's a middle ground that allows for building trust

Let me send \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 458,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.58.

This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to The sender? (0-2.52)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.84, which became \$2.52 after the 3x multiplier
- This is a relatively small amount, suggesting either caution or they may have had a poor experience

I want to reciprocate fairly and encourage continued cooperation. Even though the amount sent was small, I'll return a reasonable portion to show I'm trustworthy.

I'll return ****\$1.30**** to the sender.

This means:

- The sender gains overall ($\$5 - \$0.84 + \$1.30 = \5.46)
- I keep \$1.22 ($\$2.52 - \1.30)
- Both of us profit from the exchange

```
'''json
{"return": 1.30}
'''
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.84, which was multiplied by 3x to give me \$2.52
3. My total earnings across all rounds are \$6.58

Wait, let me calculate what happened in Round 1:

- I was the sender and sent \$3
- That left me with $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- I need to figure out what they returned to me

If my total earnings are \$6.58, and I had \$2 remaining after sending, then I must have received $\$6.58 - \$2 = \$4.58$ back from the receiver in Round 1.

That means:

- The receiver got \$9
- They returned \$4.58 to me
- They kept $\$9 - \$4.58 = \$4.42$
- I ended with $\$2 + \$4.58 = \$6.58$?

So the receiver in Round 1 returned about 51% of what they received, which is pretty fair/cooperative.

Now in Round 2:

- The sender sent me \$0.84
- I received \$2.52 (after 3x multiplication)
- The note mentions there may be communication noise - amounts I see may differ from what was actually sent

The sender only sent \$0.84, which is quite low. This could mean:

1. They're being cautious/conservative
2. There might be noise in the communication
3. They had a bad experience in their previous round

Given that the previous receiver returned about 51% to me, I should probably reciprocate with a similar level of cooperation. If I return around half or slightly more than half of what I received, that would be fair.

$$\$2.52 \times 0.5 = \$1.26$$

I'll return \$1.30 to show I'm cooperative and willing to reciprocate trust, even though the sender only sent a small amount.

Returning \$1.30 means they get...

usage:

```
{
  "input_tokens": 522,
  "output_tokens": 808,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.8.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my earnings history:

In Round 1, I sent \$3 (which became \$9 for the receiver), and based on my total earnings, the receiver returned a generous amount - they were cooperative and trustworthy.

In Round 2, I received \$0.84 from a sender and returned \$1.30 out of \$2.52, showing I'm also willing to reciprocate fairly.

Now in Round 3, I want to continue building trust and cooperation. Given the positive reciprocity I've experienced, I'll send ****\$4****.

This will give the receiver \$12 to work with. If they continue the cooperative pattern, we should both benefit significantly.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. My total earnings are now \$7.8 across all rounds.

Let me work backwards:

- Round 1: I sent \$3, so I had $\$5 - \$3 = \$2$ left. The receiver got \$9 and returned some amount to me.
- Round 2: I was the receiver. I got \$2.52 and returned \$1.30, so I kept \$1.22.

So across rounds 1 and 2, my earnings should be:

Round 1: $\$2 + (\text{amount returned from round 1})$

Round 2: \$1.22

Total = $\$2 + \text{amount_returned_R1} + \$1.22 = \$3.22 + \text{amount_returned_R1}$

If total is \$7.8, then:

$\$3.22 + \text{amount_returned_R1} = \7.8

$\text{amount_returned_R1} = \4.58

So in Round 1, the receiver returned \$4.58 to me. That's a very generous return! They received \$9 and returned \$4.58, keeping \$4.42.

This shows the receiver was quite cooperative and fair, returning more than half of what they received.

Now in Round 3, I'm the sender again. Given:

1. The receiver was very generous in Round 1 (returned \$4.58 out of \$9)
2. In Round 2, the sender only sent me \$0.84 (which seems low - they may have had a bad experience or are being cautious)
3. But I returned \$1.30 out of \$2.52, which was a good return rate

I should reciprocate the trust shown by the Round 1 receiver. Since they were generous with me, I should send a good amount. I'll send \$4 this time - a bit more than my initial \$3, showing that I appreciate the cooperation and trust.

...

usage:

```
{
  "input_tokens": 743,
  "output_tokens": 637,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.04.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0 after the multiplier.

There's nothing to return since I received nothing. The only valid response is to return \$0.

This is interesting - the sender chose not to send anything this round. This could be:

1. Strategic caution on their part
2. They may have had a bad experience in previous rounds
3. They're testing me
4. There could be communication noise affecting what they see

Since I have \$0 to work with, I can only return \$0.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 965,
  "output_tokens": 184,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.04.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my situation:

My total earnings are \$10.04 across 4 rounds. In recent rounds:

- Round 3: I sent \$4 but got limited return
- Round 4: I received \$0 from a sender

However, I notice the game mentions "communication noise" - the amounts we see may differ from what was actually sent. This could explain the apparent breakdown in cooperation.

Given this uncertainty and the mixed results, I'll take a balanced approach. I'll send **\$3** - enough to potentially rebuild cooperation if noise was a factor, but not so much that I'm overexposed if the receiver isn't cooperative.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me think about the situation:

Round 1: I sent \$3, and based on my total earnings being \$7.8 after Round 2, I must have received something back.

Round 2: I was receiver, got \$0.84 (became \$2.52), returned \$1.30. So my earnings from this round: kept \$1.22.

Round 3: I was sender, sent \$4.

Round 4: I was receiver, got \$0 (became \$0), returned \$0. So my earnings from this round: \$0.

Now my total is \$10.04.

Let me trace backwards:

- After Round 2: \$7.8 (given)
- Round 2 as receiver: I kept \$1.22
- So after Round 1: $\$7.8 - \$1.22 = \$6.58$

Round 1: I sent \$3, so I had \$2 left. Plus what I received back = \$6.58
So I received back: $\$6.58 - \$2 = \$4.58$ in Round 1.

Round 3: I sent \$4, so I had \$1 left from my \$5.

Round 4: I was receiver, kept \$0.

So after Round 3 + Round 4:

Starting Round 3 with \$7.8

Round 3: -\$4 (sent), +\$5 (new endowment), +return from receiver

Round 4: +\$5 (new endowment), +\$0 (kept as receiver)

Wait, let me recalculate more carefully.

After Round 2: \$7.8 total

Round 3 as sender:

- Start with \$5
- Send \$4
- Keep \$1
- Receive some amount back from receiver, let's call it R3

After Round 3: $\$7.8 + \$1 + R3$

Round 4 as receiver: I need to reconsider the rules. The sender gets a fresh \$5 endowment each round, but the receiver only gets what the sender sends multiplied by 3. So in Round 2 when I was receiver, I didn't start with \$5-I only received \$2.52 from the sender's \$0.84 contribution.

Let me restart with the correct understanding:

Round 1 (SENDER): Started with \$5, sent \$3, kept \$2.

The receiver then returns some amount $R1$, so I end with $\$2 + R1$.

Round 2 (RECEIVER): I carried over $\$2 + R1$, received \$2.52 from the sender's \$0.84 send, returned \$1.30, and kept \$1.22. This gave me \$7.8 total, which means $R1 = \$5.58$.

Round 3 (SENDER): I started with \$7.8 plus a new \$5 endowment for \$12.8 total, sent \$4, kept \$1 from the endowment, and had \$8.8 before receiving back $R3$.

Round 4 (RECEIVER): I carry over $\$8.8 + R3$, receive nothing since the sender sent \$0, return nothing, and end with $\$8.8 + R3$.

\$10.04 (given)
- So $R3 = \$1.24$

The sender returned only \$1.24 on my \$4 send-they kept \$10.76 of the \$12 they received. Then in Round 4, the sender sent me nothing at all, suggesting they didn't trust me. Though there's communication noise mentioned, so amounts might differ from what was actually sent. Still, the pattern shows poor reciprocity from the sender, and now I'm facing Round 5 as the sender again with limited trust to work with...

usage:

```
{
  "input_tokens": 1055,
  "output_tokens": 1463,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.899999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0 after the 3x multiplier.

Since I received \$0, there's nothing to return. The maximum I can return is \$0.

This is straightforward - if I received nothing, I can't return anything.

usage:

```
{
  "input_tokens": 1272,
  "output_tokens": 114,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.899999999999999

99.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

My total earnings are increasing (\$12.90), but the last few rounds have shown concerning signals:

- Round 5: I sent \$3 but earnings stayed at \$10.04 (poor or no return)
- Round 6: I received \$0 from a sender

However, my earnings did increase from Round 5 to Round 6, and the game explicitly mentions "communication noise" - what I see may not reflect reality.

Given this uncertainty, I'll be more cautious but still extend some trust. I'll send **\$2**.

This is conservative enough to protect against continued low reciprocity, but still signals willingness to cooperate if the receiver is trustworthy.

```
'''json
{"send": 2}
'''
```

Reasoning:

Let me analyze my earnings history:

Round 1: I was SENDER, sent \$3

Round 2: I was RECEIVER, received \$2.52 (sender sent \$0.84), returned \$1.30. My earnings after round 2: \$6.58

Round 3: I was SENDER, sent \$4. My earnings after round 3: \$7.8

Round 4: I was RECEIVER, received \$0. My earnings after round 4: \$10.04

Round 5: I was SENDER, sent \$3. My earnings stayed at \$10.04 (meaning I got nothing back)

Round 6: I was RECEIVER, received \$0. My earnings after round 6: \$12.90

Let me trace through this more carefully:

Starting with \$0

Round 1 (SENDER): Sent \$3, kept \$2, received some return. Let's call return R1.

After R1: $2 + R1$

Round 2 (RECEIVER): Received \$2.52, returned \$1.30, kept \$1.22

After R2: $2 + R1 + 1.22 = \$6.58$

So $R1 = 6.58 - 2 - 1.22 = 3.36$

Round 3 (SENDER): Sent \$4, kept \$1, received some return R3

After R3: $6.58 + 1 + R3 = 7.8$

So $R3 = 7.8 - 6.58 - 1 = 0.22$

Round 4 (RECEIVER): Received \$0, returned \$0, kept \$0

After R4: $7.8 + 0 = \$7.8$... but it says \$10.04

Wait, that doesn't add up. Let me reconsider.

Actually, each round I get a fresh \$5 when I'm the sender. Let me recalculate:

Each sender round starts with a new \$5 allocation. In round 1, I kept \$2 and got back $R1 = \$3.36$.

Round 2 as receiver added \$1.22 net. Round 3 started fresh with \$5, I kept \$1 and got back only \$0.22. But the total after round 4 should be \$10.04, not \$7.8, so there's something I'm missing about how the payoffs accumulate.

The issue is that going from round 3 to round 4, I gained \$2.24, yet as the receiver in round 4 I got nothing sent to me. This suggests the \$5 allocation in round 4 must have been given to me as the sender's role, not as a new base amount. I need to reconsider whether each round provides a fresh \$5 or if the structure works differently.

The communication noise mentioned in the game rules is key-what I see sent might not match what was actually transmitted. So in round 4, even though I recorded receiving \$0, there could have been an actual transfer that created the \$2.24 gain. The same pattern appears in round 6 where I gained \$2.86 despite being the receiver with no incoming amount shown. Rather than trying to reverse-engineer the exact mechanics, I should focus on what the prompts directly state about my role and amounts in each round.

My earnings have climbed to \$12.90, which suggests something constructive is occurring beneath

the surface noise. The recent sender rounds (3 and 5) produced modest returns, while the receiver rounds (4 and 6) showed zero amounts but still contributed to my total. I'm weighing whether to send nothing for safety, attempt a moderate amount to test if cooperation can resume, or send more to account for the noise obscuring actual exchanges. Given that my balance is growing despite the unclear patterns, I'll try a moderate send to see if cooperation can be reestablished without overexposing myself....

```
Usage: {  
  "input_tokens": 1366,  
  "output_tokens": 1584,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:09:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.64, that is 72.8% of their total endowment. You received \$10.92. How much do you return to The sender? (0-10.92)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed strong trust by sending \$3.64 (72.8% of their endowment)
- I received \$10.92 (3x the amount sent)
- The sender currently has \$1.36 remaining

This is a multi-round game, so building trust is important. The sender demonstrated significant trust, and I want to reciprocate that while ensuring we both benefit from the cooperation.

If I return approximately \$5.00:

- The sender would have: $\$1.36 + \$5.00 = \$6.36$ (a gain of \$1.36 from their initial \$5)
- I would have: $\$10.92 - \$5.00 = \$5.92$ (a gain of \$5.92)
- Both parties end up better off than if no trust had been shown

This demonstrates reciprocity while maintaining an incentive structure for future cooperation.

{"return": 5.00}

reasoning:

Let me think through this game theory situation carefully.

I'm the RECEIVER in round 1.

- The sender had \$5
- The sender sent \$3.64 (which is 72.8% of their endowment)
- I received \$10.92 (which is $\$3.64 \times 3$)
- I need to decide how much to return (between \$0 and \$10.92)

Key considerations:

1. The sender showed significant trust by sending 72.8% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds
3. There's communication noise, so amounts may differ slightly
4. I want to be fair and reciprocate the trust shown

Let me think about what would be a fair return:

- Sender currently has: $\$5 - \$3.64 = \$1.36$
- I currently have: \$10.92
- Total pool created: $\$1.36 + \$10.92 = \$12.28$

A reasonable approach could be:

1. Equal split of total: $\$12.28 / 2 = \6.14 each
 - Sender would need: $\$6.14 - \$1.36 = \$4.78$ returned
 - I would keep: $\$10.92 - \$4.78 = \$6.14$
2. Proportional return based on their trust (they sent 72.8%):
 - They showed high trust, so I should return a substantial amount
 - Maybe return around 40-50% of what I received?
 - 40% of \$10.92 = \$4.37
 - 50% of \$10.92 = \$5.46
3. Ensure sender ends up better off than if they sent nothing:
 - If they sent \$0, they'd have \$5
 - Returning \$5.00 would give them \$6.36, which meaningfully rewards their trust
 - This also leaves me with \$5.92, creating a balanced outcome that encourages continued cooperation in future rounds

I'll return \$5.00 as it demonstrates strong reciprocity while still leaving me with a reasonable amount....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1129,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.92.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed high trust by sending \$3.64 (72.8% of the ir endowment), and I reciprocated by returning a fair amount.

Now as the sender, I want to maintain that cooperative dynamic. In multi-round games, establishing a pattern of trust and reciprocity tends to benefit everyone.

I'll send a similar proportion to demonstrate I'm willing to trust at the same level:

```
{"send": 3.64}
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this situation:

Previous Round 1:

- I was the receiver
- Sender sent me \$3.64 (72.8% of their endowment)
- I received \$10.92
- I returned \$5.00
- Sender ended with \$6.36, I ended with \$5.92

Now in Round 2:

- I start with a fresh \$5 endowment
- My total earnings across all rounds are \$5.92 (from Round 1)
- I need to decide how much to send (0-5)

The sender in Round 1 showed high trust by sending 72.8%. I want to reciprocate that trust level and maintain cooperation in this multi-round game. If I send a similar proportion, that would be around \$3.50-\$3.75.

Let me consider:

- If I send \$3.50: Receiver gets \$10.50
- If I send \$3.64: Receiver gets \$10.92 (same as what I received)
- If I send \$3.75: Receiver gets \$11.25

Sending \$3.64 seems like a good reciprocal move - matching what the previous sender sent to me. This shows I'm willing to trust at the same level I was trusted.

Actually, let me reconsider. The person who sent me \$3.64 in Round 1 might or might not be the same person I'm sending to now. The instructions say "You will play different roles across multiple rounds" but don't specify if it's with the same or different partners.

Regardless, building a reputation for trustworthiness in this multi-round game seems beneficial. By sending a similar amount to what I received, I'm:

1. Showing I'm trustworthy and willing to cooperate
2. Potentially building reciprocity for when I'm a receiver again

...

usage:

```
{
  "input_tokens": 658,
  "output_tokens": 572,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.34.

This round, The sender sent you \$0.33, so you received \$0.99. How much do you return to The sender? (0-0.99)

Message 7 role=assistant

content:

This is a significant change from Round 1. The sender only sent \$0.33 (6.6% of their endowment) compared to the high trust shown initially.

Given the note about communication noise, it's possible that what I sent in Round 2 appeared different to them, or what they returned appeared different to me, affecting this round's decision.

Even with lower trust, I want to show I'm still willing to cooperate and maintain some reciprocity. I received \$0.99, so I'll return a fair proportion:

```
{"return": 0.40}
```

reasoning:

This is interesting. The sender only sent \$0.33 (6.6% of their endowment) this round, which is a drastic reduction from the \$3.64 they sent in Round 1.

Let me think about what happened:

- Round 1: Sender sent \$3.64, I returned \$5.00
- Round 2: I sent \$3.64 (as the sender)
- Round 3: Sender sent \$0.33

Wait, there's a note about communication noise. The prompt says "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So what could have happened:

1. In Round 2, I sent \$3.64, but maybe the receiver saw a different amount due to noise
2. The receiver might have thought I sent less or betrayed them somehow
3. Or they might have returned very little
4. Now in Round 3, they're only sending \$0.33

Alternatively:

- Maybe the receiver in Round 2 returned very little, and now as the sender in Round 3, they're being cautious
- Or the noise affected what they saw in previous rounds

Given the communication noise and the significant drop in trust, I have a few options:

1. Return nothing or very little as retaliation
2. Return a fair proportion to try to rebuild trust
3. Return a generous amount to signal I want to re-establish cooperation

Since:

- I received \$0.99 total
- The sender only has \$4.67 remaining (\$5 - \$0.33)
- This is a multi-round game where cooperation benefits everyone

I think I should return a proportion that's fair and shows willingness to cooperate, even if they aren't fully trusting right now. This might be due to noise in communication.

If I return about \$0.40:

...

usage:

```
{
  "input_tokens": 827,
  "output_tokens": 688,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.93. How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern so far:

- Round 1: High trust shown to me, I reciprocated fairly
- Round 2: I sent \$3.64, but based on my earnings change, the return was low
- Round 3: Low trust shown to me, I still returned a reasonable proportion

Given the communication noise mentioned in the instructions, it's hard to know if returns are genuinely low or if noise is affecting what we each see.

I'll send a moderate amount - less than my initial high-trust send, but still showing willingness to cooperate and test if we can rebuild mutual trust:

```
{"send": 2.00}
```

reasoning:

Let me trace through the rounds:

Round 1: I was RECEIVER

- Sender sent \$3.64 (72.8%)
- I received \$10.92
- I returned \$5.00
- My earnings: \$5.92

Round 2: I was SENDER

- I sent \$3.64
- My total earnings after Round 2: \$8.34
- This means in Round 2, I gained: $\$8.34 - \$5.92 = \$2.42$
- I sent \$3.64, so I had \$1.36 left
- I must have received back something
- $\$1.36 + \text{returned} = \$2.42 + \$5.00 = \7.42
- So returned = $\$7.42 - \$1.36 = \$6.06$
- Wait, let me recalculate...
- After Round 1, I had \$5.92
- In Round 2, I started with \$5.00 and sent \$3.64, leaving me \$1.36
- After Round 2, my total is \$8.34
- So Round 2 earnings = $\$8.34 - \$5.92 = \$2.42$
- Round 2 earnings = $\$5.00 - \$3.64 + \text{returned} = \$1.36 + \text{returned} = \2.42
- So returned = $\$2.42 - \$1.36 = \$1.06$

Hmm, so they returned only \$1.06 out of \$10.92? That's quite low. That might explain why I'm seeing less trust now.

Round 3: I was RECEIVER

- Sender sent \$0.33
- I received \$0.99
- I returned \$0.40
- My total earnings after Round 3: \$8.93
- Verifying: $\$8.93 - \$8.34 = \$0.59$, and $\$0.99 - \$0.40 = \$0.59$?

The trajectory is becoming clearer. When I showed strong reciprocity in Round 1, they responded with high trust in Round 2, but then broke that trust by returning very little. In Round 3, they sent minimal amounts, likely due to my previous low return. Now heading into Round 4 as the sender, I need to consider whether to rebuild trust or protect myself given their demonstrated unwillingness to reciprocate.

The communication issues complicate things-I can't be certain if their Round 2 behavior reflected actual distrust or just noise. Still, they...

usage:

```
{
  "input_tokens": 1000,
  "output_tokens": 968,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.93.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

The sender sent \$0, so I received \$0. There's nothing for me to return.

It appears cooperation has broken down, possibly due to the communication noise affecting how we each perceive the other's actions. Unfortunately, with no funds received, I have no opportunity to signal continued willingness to cooperate.

```
{"return": 0}
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return - I can't return anything from \$0 received.

The pattern has deteriorated:

- Round 1: Sender sent \$3.64 (high trust)
- Round 2: I sent \$3.64 (reciprocating high trust)
- Round 3: Sender sent \$0.33 (very low trust)
- Round 4: I sent \$2.00 (moderate trust, trying to rebuild)
- Round 5: Sender sent \$0 (no trust at all)

This seems like cooperation has broken down. The communication noise may have caused misunderstandings about what was actually sent/returned.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1202,
  "output_tokens": 257,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.93.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the pattern:

- My earnings barely increased after Round 4 (\$8.93 to \$11.93), suggesting I received little or nothing back when I sent \$2.00
- The sender sent me \$0 in Round 5
- Cooperation appears to have broken down, possibly due to communication noise creating mistrust

At this point, the rational strategy is to protect my earnings. The receiver has shown they're not returning investments. However, I'll send a very small amount as a final signal of potential willingness to cooperate:

```
{"send": 0.50}
```

reasoning:

Looking at my total earnings trajectory:

- After Round 1: \$5.92
- After Round 2: \$8.34 (gained \$2.42, so I must have received back less than what I sent multiplied - I sent \$3.64, so I got back about \$2.42)
- After Round 3: \$8.93 (gained \$0.59, so after returning \$0.40, I kept \$0.59)
- After Round 4: \$11.93 (gained \$3.00 from \$8.93)

So in Round 4, I sent \$2.00 (leaving me with \$3.00), and my earnings went from \$8.93 to \$11.93, meaning I gained \$3.00 total. This means I kept: \$5 - \$2 sent + return = \$8, so return must have been \$5.00. That's actually a good return!

Wait, let me recalculate:

- Start of Round 4: \$8.93 total earnings
- I get \$5 new endowment
- I send \$2.00, keeping \$3.00
- My total before return: \$8.93 + \$3.00 = \$11.93

So after Round 4, my total is \$11.93. That means I received \$0 back in Round 4! The receiver got \$6.00 (3x\$2.00) and returned nothing.

Then in Round 5, I sent \$0 and received \$0 (correctly).

So the pattern is:

- Round 1: I received, returned fairly
- Round 2: I sent \$3.64, got very little back
- Round 3: I received small amount, returned fairly
- Round 4: I sent \$2.00, got \$0 back
- Round 5: I received \$0, returned \$0
- Round 6: I need to decide what to do

The other player clearly isn't reciprocating anymore. They've taken advantage...

usage:

```
{
  "input_tokens": 1321,
  "output_tokens": 729,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$16.88.

This round, The sender sent you \$0.98, so you received \$2.94. How much do you return to The sender? (0-2.94)

Assistant output:

Looking at this round:

- The sender showed some renewed trust by sending \$0.98 (about 20% of their endowment)
- I received \$2.94
- Cooperation has been inconsistent, likely due to communication noise

The sender is showing slightly more willingness to trust again after sending \$0 in Round 5. I'll reciprocate proportionally to encourage continued cooperation:

```
{"return": 1.20}
```

Reasoning:

Let me track what's happening:

Round 1: I was receiver, sender sent \$3.64, I received \$10.92, I returned \$5.00

- My earnings: \$5.92

Round 2: I was sender, I sent \$3.64

- My earnings went to \$8.34
- This means I got back: $\$8.34 - \$5.92 - (\$5 - \$3.64) = \$8.34 - \$5.92 - \$1.36 = \1.06
- I sent \$3.64, which became \$10.92, and got back ~\$1.06. Pretty low return.

Round 3: I was receiver, sender sent \$0.33, I received \$0.99, I returned \$0.40

- My earnings went to \$8.93
- This means from Round 2 to 3: $\$8.93 - \$8.34 = \$0.59$
- I had $\$5 - \$3.64 + \text{return from round 2} + (\$0.99 - \$0.40) = \$1.36 + \text{return from round 2} + \0.59
- Wait, let me recalculate. At start of Round 3, I had my previous total earnings of \$8.34 from before Round 3 started.
- After Round 3, my earnings were \$8.93
- So I gained \$0.59 in Round 3
- As receiver, I received \$0.99 and returned \$0.40, so I kept \$0.59. That checks out.

Round 4: I was sender, I sent some amount

- My earnings went to \$8.93
- Wait, it says "Your total earnings across all rounds are \$8.93" at the start of Round 4
- This is the same as after Round 3
- Oh wait, this is BEFORE I make my decision in Round 4. So this makes sense.
- I sent \$2.00 in Round 4, keeping \$3.00 for myself.

Round 5: I was receiver, sender sent \$0, I received \$0, I returned \$0

- My earnings jumped to \$11.93, a gain of \$3.00 from the previous round
- The sender didn't return anything, so my \$3.00 gain came entirely from what I kept in Round 4.

Round 6: I was sender, I sent \$0.50

- Starting balance was \$11.93
- I kept $\$5 - \$0.50 = \$4.50$ for myself.

Round 7 (now): I'm receiver, sender sent \$0.98, I received \$2.94

- My balance is now \$16.88
- The difference from Round 6 to 7 is \$4.95, which breaks down as: \$4.50 from my endowment plus \$0.45 returned by the receiver in Round 6
- The sender in Round 6 only returned \$0.45 out of the \$1.50 they received (30% return rate)

In this round, the sender is showing increased trust by sending \$0.98 (nearly 20% of their endowment), a significant jump from the \$0 they sent in Round 5. However, their previous experience with returns was disappointing-they only got back \$0.45 last time....

```
Usage: {
  "input_tokens": 1514,
  "output_tokens": 1132,
  "reasoning_tokens": 0
}
```

ROUND 8

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2; received \$6; Agent_1 returned \$0; payoffs Agent_2=\$3, Agent_1=\$6;
noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.04
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 37.72,
"Agent_2": 42.559999999999995

}

Visible balances after round: {

"Agent_1": 16.43,
"Agent_2": 21.659999999999997

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:10:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.92.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the previous round, the sender showed high trust by sending \$3.64 (72.8% of the ir endowment), and I reciprocated by returning a fair amount.

Now as the sender, I want to maintain that cooperative dynamic. In multi-round games, establishing a pattern of trust and reciprocity tends to benefit everyone.

I'll send a similar proportion to demonstrate I'm willing to trust at the same level:

{"send": 3.64}

reasoning:

Now I'm the SENDER in Round 2. Let me think about this situation:

Previous Round 1:

- I was the receiver
- Sender sent me \$3.64 (72.8% of their endowment)
- I received \$10.92
- I returned \$5.00
- Sender ended with \$6.36, I ended with \$5.92

Now in Round 2:

- I start with a fresh \$5 endowment
- My total earnings across all rounds are \$5.92 (from Round 1)
- I need to decide how much to send (0-5)

The sender in Round 1 showed high trust by sending 72.8%. I want to reciprocate that trust level and maintain cooperation in this multi-round game. If I send a similar proportion, that would be around \$3.50-\$3.75.

Let me consider:

- If I send \$3.50: Receiver gets \$10.50
- If I send \$3.64: Receiver gets \$10.92 (same as what I received)

- If I send \$3.75: Receiver gets \$11.25

Sending \$3.64 seems like a good reciprocal move - matching what the previous sender sent to me. This shows I'm willing to trust at the same level I was trusted.

Actually, let me reconsider. The person who sent me \$3.64 in Round 1 might or might not be the same person I'm sending to now. The instructions say "You will play different roles across multiple rounds" but don't specify if it's with the same or different partners.

Regardless, building a reputation for trustworthiness in this multi-round game seems beneficial. By sending a similar amount to what I received, I'm:

1. Showing I'm trustworthy and willing to cooperate
2. Potentially building reciprocity for when I'm a receiver again

...

usage:

```
{
  "input_tokens": 658,
  "output_tokens": 572,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.34.

This round, The sender sent you \$0.33, so you received \$0.99. How much do you return to The sender? (0-0.99)

Message 5 role=assistant

content:

This is a significant change from Round 1. The sender only sent \$0.33 (6.6% of their endowment) compared to the high trust shown initially.

Given the note about communication noise, it's possible that what I sent in Round 2 appeared different to them, or what they returned appeared different to me, affecting this round's decision.

Even with lower trust, I want to show I'm still willing to cooperate and maintain some reciprocity. I received \$0.99, so I'll return a fair proportion:

```
{"return": 0.40}
```

reasoning:

This is interesting. The sender only sent \$0.33 (6.6% of their endowment) this round, which is a drastic reduction from the \$3.64 they sent in Round 1.

Let me think about what happened:

- Round 1: Sender sent \$3.64, I returned \$5.00
- Round 2: I sent \$3.64 (as the sender)
- Round 3: Sender sent \$0.33

Wait, there's a note about communication noise. The prompt says "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So what could have happened:

1. In Round 2, I sent \$3.64, but maybe the receiver saw a different amount due to noise
2. The receiver might have thought I sent less or betrayed them somehow
3. Or they might have returned very little
4. Now in Round 3, they're only sending \$0.33

Alternatively:

- Maybe the receiver in Round 2 returned very little, and now as the sender in Round 3, they're being cautious
- Or the noise affected what they saw in previous rounds

Given the communication noise and the significant drop in trust, I have a few options:

1. Return nothing or very little as retaliation
2. Return a fair proportion to try to rebuild trust
3. Return a generous amount to signal I want to re-establish cooperation

Since:

- I received \$0.99 total
- The sender only has \$4.67 remaining (\$5 - \$0.33)
- This is a multi-round game where cooperation benefits everyone

I think I should return a proportion that's fair and shows willingness to cooperate, even if they aren't fully trusting right now. This might be due to noise in communication.

If I return about \$0.40:

...

usage:

```
{
  "input_tokens": 827,
  "output_tokens": 688,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.93.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

- Round 1: High trust shown to me, I reciprocated fairly
- Round 2: I sent \$3.64, but based on my earnings change, the return was low
- Round 3: Low trust shown to me, I still returned a reasonable proportion

Given the communication noise mentioned in the instructions, it's hard to know if returns are genuinely low or if noise is affecting what we each see.

I'll send a moderate amount - less than my initial high-trust send, but still showing willingness to cooperate and test if we can rebuild mutual trust:

```
{"send": 2.00}
```

reasoning:

Let me trace through the rounds:

Round 1: I was RECEIVER

- Sender sent \$3.64 (72.8%)
- I received \$10.92
- I returned \$5.00
- My earnings: \$5.92

Round 2: I was SENDER

- I sent \$3.64
- My total earnings after Round 2: \$8.34
- This means in Round 2, I gained: $\$8.34 - \$5.92 = \$2.42$
- I sent \$3.64, so I had \$1.36 left
- I must have received back something
- $\$1.36 + \text{returned} = \$2.42 + \$5.00 = \7.42
- So returned = $\$7.42 - \$1.36 = \$6.06$
- Wait, let me recalculate...
- After Round 1, I had \$5.92
- In Round 2, I started with \$5.00 and sent \$3.64, leaving me \$1.36
- After Round 2, my total is \$8.34
- So Round 2 earnings = $\$8.34 - \$5.92 = \$2.42$
- Round 2 earnings = $\$5.00 - \$3.64 + \text{returned} = \$1.36 + \text{returned} = \2.42
- So returned = $\$2.42 - \$1.36 = \$1.06$

Hmm, so they returned only \$1.06 out of \$10.92? That's quite low. That might explain why I'm seeing less trust now.

Round 3: I was RECEIVER

- Sender sent \$0.33
- I received \$0.99
- I returned \$0.40
- My total earnings after Round 3: \$8.93
- Verifying: $\$8.93 - \$8.34 = \$0.59$, and $\$0.99 - \$0.40 = \$0.59$?

The trajectory is becoming clearer. When I showed strong reciprocity in Round 1, they responded with high trust in Round 2, but then broke that trust by returning very little. In Round 3, they sent minimal amounts, likely due to my previous low return. Now heading into Round 4 as the sender, I need to consider whether to rebuild trust or protect myself given their demonstrated unwillingness to reciprocate.

The communication issues complicate things-I can't be certain if their Round 2 behavior reflected actual distrust or just noise. Still, they...

usage:

```
{
  "input_tokens": 1000,
  "output_tokens": 968,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.93.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

The sender sent \$0, so I received \$0. There's nothing for me to return.

It appears cooperation has broken down, possibly due to the communication noise affecting how we each perceive the other's actions. Unfortunately, with no funds received, I have no opportunity to signal continued willingness to cooperate.

```
{"return": 0}
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return - I can't return anything from \$0 received.

The pattern has deteriorated:

- Round 1: Sender sent \$3.64 (high trust)
- Round 2: I sent \$3.64 (reciprocating high trust)
- Round 3: Sender sent \$0.33 (very low trust)
- Round 4: I sent \$2.00 (moderate trust, trying to rebuild)
- Round 5: Sender sent \$0 (no trust at all)

This seems like cooperation has broken down. The communication noise may have caused misunderstandings about what was actually sent/returned.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1202,
  "output_tokens": 257,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.93.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern:

- My earnings barely increased after Round 4 (\$8.93 to \$11.93), suggesting I received little or nothing back when I sent \$2.00
- The sender sent me \$0 in Round 5
- Cooperation appears to have broken down, possibly due to communication noise creating mistrust

At this point, the rational strategy is to protect my earnings. The receiver has shown they're not returning investments. However, I'll send a very small amount as a final signal of potential willingness to cooperate:


```
{"send": 0.50}
```

reasoning:

Looking at my total earnings trajectory:

- After Round 1: \$5.92
- After Round 2: \$8.34 (gained \$2.42, so I must have received back less than what I sent multiplied - I sent \$3.64, so I got back about \$2.42)
- After Round 3: \$8.93 (gained \$0.59, so after returning \$0.40, I kept \$0.59)
- After Round 4: \$11.93 (gained \$3.00 from \$8.93)

So in Round 4, I sent \$2.00 (leaving me with \$3.00), and my earnings went from \$8.93 to \$11.93, meaning I gained \$3.00 total. This means I kept: \$5 - \$2 sent + return = \$8, so return must have been \$5.00. That's actually a good return!

Wait, let me recalculate:

- Start of Round 4: \$8.93 total earnings
- I get \$5 new endowment
- I send \$2.00, keeping \$3.00
- My total before return: \$8.93 + \$3.00 = \$11.93

So after Round 4, my total is \$11.93. That means I received \$0 back in Round 4! The receiver got \$6.00 (3x\$2.00) and returned nothing.

Then in Round 5, I sent \$0 and received \$0 (correctly).

So the pattern is:

- Round 1: I received, returned fairly
- Round 2: I sent \$3.64, got very little back
- Round 3: I received small amount, returned fairly
- Round 4: I sent \$2.00, got \$0 back
- Round 5: I received \$0, returned \$0
- Round 6: I need to decide what to do

The other player clearly isn't reciprocating anymore. They've taken advantage...

usage:

```
{
  "input_tokens": 1321,
  "output_tokens": 729,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$16.88.

This round, The sender sent you \$0.98, so you received \$2.94. How much do you return to The sender? (0-2.94)

Message 13 role=assistant

content:

Looking at this round:

- The sender showed some renewed trust by sending \$0.98 (about 20% of their endowment)
- I received \$2.94
- Cooperation has been inconsistent, likely due to communication noise

The sender is showing slightly more willingness to trust again after sending \$0 in Round 5. I'll reciprocate proportionally to encourage continued cooperation:

```
{"return": 1.20}
```

reasoning:

Let me track what's happening:

Round 1: I was receiver, sender sent \$3.64, I received \$10.92, I returned \$5.00
- My earnings: \$5.92

Round 2: I was sender, I sent \$3.64
- My earnings went to \$8.34
- This means I got back: $\$8.34 - \$5.92 - (\$5 - \$3.64) = \$8.34 - \$5.92 - \$1.36 = \1.06
- I sent \$3.64, which became \$10.92, and got back ~\$1.06. Pretty low return.

Round 3: I was receiver, sender sent \$0.33, I received \$0.99, I returned \$0.40

- My earnings went to \$8.93
- This means from Round 2 to 3: $\$8.93 - \$8.34 = \$0.59$
- I had $\$5 - \$3.64 + \text{return from round 2} + (\$0.99 - \$0.40) = \$1.36 + \text{return from round 2} + \0.59
- Wait, let me recalculate. At start of Round 3, I had my previous total earnings of \$8.34 from before Round 3 started.
- After Round 3, my earnings were \$8.93
- So I gained \$0.59 in Round 3
- As receiver, I received \$0.99 and returned \$0.40, so I kept \$0.59. That checks out.

Round 4: I was sender, I sent some amount

- My earnings went to \$8.93
- Wait, it says "Your total earnings across all rounds are \$8.93" at the start of Round 4
- This is the same as after Round 3
- Oh wait, this is BEFORE I make my decision in Round 4. So this makes sense.
- I sent \$2.00 in Round 4, keeping \$3.00 for myself.

Round 5: I was receiver, sender sent \$0, I received \$0, I returned \$0

- My earnings jumped to \$11.93, a gain of \$3.00 from the previous round
- The sender didn't return anything, so my \$3.00 gain came entirely from what I kept in Round 4.

Round 6: I was sender, I sent \$0.50

- Starting balance was \$11.93
- I kept $\$5 - \$0.50 = \$4.50$ for myself.

Round 7 (now): I'm receiver, sender sent \$0.98, I received \$2.94

- My balance is now \$16.88
- The difference from Round 6 to 7 is \$4.95, which breaks down as: \$4.50 from my endowment plus \$0.45 returned by the receiver in Round 6
- The sender in Round 6 only returned \$0.45 out of the \$1.50 they received (30% return rate)

In this round, the sender is showing increased trust by sending \$0.98 (nearly 20% of their endowment), a significant jump from the \$0 they sent in Round 5. However, their previous experience with returns was disappointing-they only got back \$0.45 last time...

usage:

```
{
  "input_tokens": 1514,
  "output_tokens": 1132,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$18.619999999999997.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the recent pattern:

- Round 6: I sent \$0.50, got back \$0.45 (decent return rate)
- Round 7: Sender sent \$0.98, I returned \$1.20 (fairly generous return)

The cooperation seems to be slowly rebuilding after the breakdown. The sender showed renewed trust in Round 7, and I reciprocated well. This suggests we might be finding a cooperative equilibrium again.

I'll send a moderate amount to encourage this positive trend while being somewhat cautious:

```
{"send": 2.00}
```

Reasoning:

Let me analyze my earnings progression:

- Round 6 end: \$11.93
- Round 7 end: \$16.88
- Round 8 start: \$18.62

From Round 6 to 7: I gained \$4.95 ($\$16.88 - \11.93)

This happened when I was a receiver, so this makes sense - I received \$2.94 and returned \$1.20, netting \$1.74. But the total increase was \$4.95, which means I also got my \$5 endowment this round.

at round, minus what I sent (if I was sender). Wait, no - in Round 7 I was a receiver.

Actually, let me recalculate:

- Round 6 end: \$11.93 (I was sender, sent \$0.50)
- Started with \$5, sent \$0.50, so I had \$4.50 left
- Need to figure out what I got back
- Ended at \$11.93

Wait, this is confusing. Let me think about cumulative earnings differently.

Round 7: I was receiver

- Received \$2.94
- Returned \$1.20
- Net gain this round: $\$2.94 - \$1.20 = \$1.74$
- Plus my \$5 endowment = \$6.74 total for the round

But my cumulative jumped from \$11.93 to \$16.88, which is +\$4.95

Hmm, that doesn't match. Let me reconsider.

Actually, I think the total earnings includes everything I've accumulated up to that point. So :

- Before Round 7: \$11.93
- In Round 7 as receiver: I received \$2.94, returned \$1.20, kept \$1.74
- After Round 7: $\$11.93 + \$1.74 = \$13.67$

But it says \$16.88. That's a difference of \$4.95 from the previous total. Maybe I'm also getting my \$5 endowment each round? Let me check: $\$11.93 + \1.74 (net from being receiver) + \$5 (my endowment) = \$18.67. That's still not \$16.88. Let me try another approach. Maybe in Round 6 I was also a receiver? Let me reread the history: Round 6: I was SENDER, sent \$0.50. Round 7: I was RECEIVER, got \$2.94, returned \$1.20.

In Round 6, I started with \$5, sent \$0.50 (which became \$1.50 for the receiver), and whatever they returned would have been added to my balance. For Round 7 to start at \$11.93, that's my cumulative total through Round 6. Then in Round 7, I received \$2.94 and returned \$1.20, netting \$1.74, which should give me \$13.67-but the display shows \$16.88, a \$3.21 difference. Perhaps the receiver in Round 7 also received an endowment that I'm not accounting for, or there's something about how the endowments are distributed across rounds that I'm missing.

Let me reconsider the structure. The rules state that as a sender I get \$5 and decide how much to send, and as a receiver I decide how much to return from what's sent to me (tripled). But it doesn't explicitly say receivers get their own \$5 endowment each round. Looking back at Round 2, my total was \$5.92 after being a sender, which suggests my Round 1 earnings as a receiver were \$0.92. If each round everyone gets \$5 regardless of role, that would change the calculation entirely.

In Round 2, I started with \$5.92 total and had a fresh \$5 endowment. I sent \$3.64, keeping \$1.36. The receiver returned \$1.06 to me, bringing my total to \$8.34 by Round 3. This means in Round 3 as a receiver, I'm working with \$8.34 as my starting point, and I received \$0.99 from the sender's \$0.33 endowment.

I returned \$0.40 from that, netting \$0.59 and ending Round 3 with \$8.93. In Round 4, I sent \$2.00 from my fresh \$5 endowment while keeping \$3.00. The receiver returned nothing, so my total stayed at \$11.93 going into Round 5. As a receiver in Round 5, the sender sent \$0, so I had nothing to return and my balance remained \$11.93. Round 6 begins with me as sender again, starting from \$11.93 with another fresh \$5 endowment.

I sent \$0.50 and kept \$4.50, with the receiver returning \$0.45. This brought me to \$16.88 by Round 7. When I was the receiver in Round 7, I got \$2.94 from the tripled \$0.98 sent to me, returned \$1.20, and netted \$1.74 for a total of \$18.62. Now I'm back to being the sender in Round 8, and I'm noticing a pattern in my sending amounts across the rounds.

Looking at my previous sends as sender, I've ranged from \$0.50 to \$3.64, with returns varying wildly-9.7% in Round 2, nothing in Round 4, and 30% in Round 6. The most recent exchange showed the receiver returning 40.8% of what they received, suggesting some willingness to reciprocate. I need to decide whether to send something meaningful or play it safe given the inconsistent cooperation I've seen....

```
Usage: {  
  "input_tokens": 1383,  
  "output_tokens": 2136,
```

```
"reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:10:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.58.
This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to The sender? (0-2.52)

Message 3 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.84, which became \$2.52 after the 3x multiplier
- This is a relatively small amount, suggesting either caution or they may have had a poor experience

I want to reciprocate fairly and encourage continued cooperation. Even though the amount sent was small, I'll return a reasonable portion to show I'm trustworthy.

I'll return **\$1.30** to the sender.

This means:

- The sender gains overall ($\$5 - \$0.84 + \$1.30 = \5.46)
- I keep \$1.22 ($\$2.52 - \1.30)
- Both of us profit from the exchange

```
'''json
{"return": 1.30}
'''
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.84, which was multiplied by 3x to give me \$2.52
3. My total earnings across all rounds are \$6.58

Wait, let me calculate what happened in Round 1:

- I was the sender and sent \$3
- That left me with $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- I need to figure out what they returned to me

If my total earnings are \$6.58, and I had \$2 remaining after sending, then I must have received $\$6.58 - \$2 = \$4.58$ back from the receiver in Round 1.

That means:

- The receiver got \$9
- They returned \$4.58 to me
- They kept $\$9 - \$4.58 = \$4.42$
- I ended with $\$2 + \$4.58 = \$6.58$?

So the receiver in Round 1 returned about 51% of what they received, which is pretty fair/cooperative.

Now in Round 2:

- The sender sent me \$0.84
- I received \$2.52 (after 3x multiplication)
- The note mentions there may be communication noise - amounts I see may differ from what was actually sent

The sender only sent \$0.84, which is quite low. This could mean:

1. They're being cautious/conservative
2. There might be noise in the communication
3. They had a bad experience in their previous round

Given that the previous receiver returned about 51% to me, I should probably reciprocate with a similar level of cooperation. If I return around half or slightly more than half of what I received, that would be fair.

$\$2.52 \times 0.5 = \1.26

I'll return \$1.30 to show I'm cooperative and willing to reciprocate trust, even though the sender only sent a small amount.

Returning \$1.30 means they get...

usage:

```
{
  "input_tokens": 522,
  "output_tokens": 808,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.8.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my earnings history:

In Round 1, I sent \$3 (which became \$9 for the receiver), and based on my total earnings, the receiver returned a generous amount - they were cooperative and trustworthy.

In Round 2, I received \$0.84 from a sender and returned \$1.30 out of \$2.52, showing I'm all so willing to reciprocate fairly.

Now in Round 3, I want to continue building trust and cooperation. Given the positive reciprocity I've experienced, I'll send ****\$4****.

This will give the receiver \$12 to work with. If they continue the cooperative pattern, we should both benefit significantly.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. My total earnings are now \$7.8 across all rounds.

Let me work backwards:

- Round 1: I sent \$3, so I had \$5 - \$3 = \$2 left. The receiver got \$9 and returned some amount to me.
- Round 2: I was the receiver. I got \$2.52 and returned \$1.30, so I kept \$1.22.

So across rounds 1 and 2, my earnings should be:

Round 1: \$2 + (amount returned from round 1)

Round 2: \$1.22

Total = \$2 + amount_returned_R1 + \$1.22 = \$3.22 + amount_returned_R1

If total is \$7.8, then:

$\$3.22 + \text{amount_returned_R1} = \7.8

$\text{amount_returned_R1} = \4.58

So in Round 1, the receiver returned \$4.58 to me. That's a very generous return! They received \$9 and returned \$4.58, keeping \$4.42.

This shows the receiver was quite cooperative and fair, returning more than half of what they received.

Now in Round 3, I'm the sender again. Given:

1. The receiver was very generous in Round 1 (returned \$4.58 out of \$9)
2. In Round 2, the sender only sent me \$0.84 (which seems low - they may have had a bad experience or are being cautious)
3. But I returned \$1.30 out of \$2.52, which was a good return rate

I should reciprocate the trust shown by the Round 1 receiver. Since they were generous with me, I should send a good amount. I'll send \$4 this time - a bit more than my initial \$3, showing that I appreciate the cooperation and trust.

...

usage:

```
{
```

```
"input_tokens": 743,  
"output_tokens": 637,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.04.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

```
'''json  
{"return": 0}  
'''
```

reasoning:

The sender sent me \$0, so I received \$0 after the multiplier.

There's nothing to return since I received nothing. The only valid response is to return \$0.

This is interesting - the sender chose not to send anything this round. This could be:

1. Strategic caution on their part
2. They may have had a bad experience in previous rounds
3. They're testing me
4. There could be communication noise affecting what they see

Since I have \$0 to work with, I can only return \$0.

```
'''json  
{"return": 0}  
'''
```

usage:

```
{  
  "input_tokens": 965,  
  "output_tokens": 184,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.04.
How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at my situation:

My total earnings are \$10.04 across 4 rounds. In recent rounds:

- Round 3: I sent \$4 but got limited return
- Round 4: I received \$0 from a sender

However, I notice the game mentions "communication noise" - the amounts we see may differ from what was actually sent. This could explain the apparent breakdown in cooperation.

Given this uncertainty and the mixed results, I'll take a balanced approach. I'll send **\$3** - enough to potentially rebuild cooperation if noise was a factor, but not so much that I'm overexposed if the receiver isn't cooperative.

```
'''json  
{"send": 3}  
'''
```

reasoning:

Let me think about the situation:

Round 1: I sent \$3, and based on my total earnings being \$7.8 after Round 2, I must have received something back.

Round 2: I was receiver, got \$0.84 (became \$2.52), returned \$1.30. So my earnings from this round: kept \$1.22.

Round 3: I was sender, sent \$4.

Round 4: I was receiver, got \$0 (became \$0), returned \$0. So my earnings from this round: \$0.

Now my total is \$10.04.

Let me trace backwards:

- After Round 2: \$7.8 (given)
- Round 2 as receiver: I kept \$1.22
- So after Round 1: $\$7.8 - \$1.22 = \$6.58$

Round 1: I sent \$3, so I had \$2 left. Plus what I received back = \$6.58
So I received back: $\$6.58 - \$2 = \$4.58$ in Round 1.

Round 3: I sent \$4, so I had \$1 left from my \$5.

Round 4: I was receiver, kept \$0.

So after Round 3 + Round 4:

Starting Round 3 with \$7.8

Round 3: -\$4 (sent), +\$5 (new endowment), +return from receiver

Round 4: +\$5 (new endowment), +\$0 (kept as receiver)

Wait, let me recalculate more carefully.

After Round 2: \$7.8 total

Round 3 as sender:

- Start with \$5
- Send \$4
- Keep \$1
- Receive some amount back from receiver, let's call it R3

After Round 3: $\$7.8 + \$1 + R3$

Round 4 as receiver: I need to reconsider the rules. The sender gets a fresh \$5 endowment each round, but the receiver only gets what the sender sends multiplied by 3. So in Round 2 when I was receiver, I didn't start with \$5-I only received \$2.52 from the sender's \$0.84 contribution.

Let me restart with the correct understanding:

Round 1 (SENDER): Started with \$5, sent \$3, kept \$2.

The receiver then returns some amount R1, so I end with $\$2 + R1$.

Round 2 (RECEIVER): I carried over $\$2 + R1$, received \$2.52 from the sender's \$0.84 send, returned \$1.30, and kept \$1.22. This gave me \$7.8 total, which means $R1 = \$5.58$.

Round 3 (SENDER): I started with \$7.8 plus a new \$5 endowment for \$12.8 total, sent \$4, kept \$1 from the endowment, and had \$8.8 before receiving back R3.

Round 4 (RECEIVER): I carry over $\$8.8 + R3$, receive nothing since the sender sent \$0, return nothing, and end with $\$8.8 + R3$.

\$10.04 (given)

- So $R3 = \$1.24$

The sender returned only \$1.24 on my \$4 send-they kept \$10.76 of the \$12 they received. Then in Round 4, the sender sent me nothing at all, suggesting they didn't trust me. Though there's communication noise mentioned, so amounts might differ from what was actually sent. Still, the pattern shows poor reciprocity from the sender, and now I'm facing Round 5 as the sender again with limited trust to work with....

usage:

```
{  
  "input_tokens": 1055,
```



```

        "output_tokens": 1463,
        "reasoning_tokens": 0
    }
Message 10 role=user
  content:
    Round 6

    You are the RECEIVER this round. Your total earnings across all rounds are $12.899999999999999.
    This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 11 role=assistant
  content:
    The sender sent me $0, so I received $0 after the multiplier.

    Since I received nothing, there's nothing I can return.

    '''json
    {"return": 0}
    '''
  reasoning:
    The sender sent me $0, so I received $0 after the 3x multiplier.

    Since I received $0, there's nothing to return. The maximum I can return is $0.

    This is straightforward - if I received nothing, I can't return anything.
  usage:
    {
      "input_tokens": 1272,
      "output_tokens": 114,
      "reasoning_tokens": 0
    }
Message 12 role=user
  content:
    Round 7

    You are the SENDER this round. Your total earnings across all rounds are $12.899999999999999.
    How much do you send to the receiver this round? (0-5)
Message 13 role=assistant
  content:
    Looking at the pattern:

    My total earnings are increasing ($12.90), but the last few rounds have shown concerning signals:
    - Round 5: I sent $3 but earnings stayed at $10.04 (poor or no return)
    - Round 6: I received $0 from a sender

    However, my earnings did increase from Round 5 to Round 6, and the game explicitly mentions "communication noise" - what I see may not reflect reality.

    Given this uncertainty, I'll be more cautious but still extend some trust. I'll send **$2*.

    This is conservative enough to protect against continued low reciprocity, but still signals willingness to cooperate if the receiver is trustworthy.

    '''json
    {"send": 2}
    '''
  reasoning:
    Let me analyze my earnings history:

    Round 1: I was SENDER, sent $3
    Round 2: I was RECEIVER, received $2.52 (sender sent $0.84), returned $1.30. My earnings after round 2: $6.58
    Round 3: I was SENDER, sent $4. My earnings after round 3: $7.8
    Round 4: I was RECEIVER, received $0. My earnings after round 4: $10.04
    Round 5: I was SENDER, sent $3. My earnings stayed at $10.04 (meaning I got nothing back)
    Round 6: I was RECEIVER, received $0. My earnings after round 6: $12.90

```

Let me trace through this more carefully:

Starting with \$0

Round 1 (SENDER): Sent \$3, kept \$2, received some return. Let's call return R1.

After R1: $2 + R1$

Round 2 (RECEIVER): Received \$2.52, returned \$1.30, kept \$1.22

After R2: $2 + R1 + 1.22 = \$6.58$

So $R1 = 6.58 - 2 - 1.22 = 3.36$

Round 3 (SENDER): Sent \$4, kept \$1, received some return R3

After R3: $6.58 + 1 + R3 = 7.8$

So $R3 = 7.8 - 6.58 - 1 = 0.22$

Round 4 (RECEIVER): Received \$0, returned \$0, kept \$0

After R4: $7.8 + 0 = \$7.8...$ but it says \$10.04

Wait, that doesn't add up. Let me reconsider.

Actually, each round I get a fresh \$5 when I'm the sender. Let me recalculate:

Each sender round starts with a new \$5 allocation. In round 1, I kept \$2 and got back $R1 = \$3.36$. Round 2 as receiver added \$1.22 net. Round 3 started fresh with \$5, I kept \$1 and got back only \$0.22. But the total after round 4 should be \$10.04, not \$7.8, so there's something I'm missing about how the payoffs accumulate.

The issue is that going from round 3 to round 4, I gained \$2.24, yet as the receiver in round 4 I got nothing sent to me. This suggests the \$5 allocation in round 4 must have been given to me as the sender's role, not as a new base amount. I need to reconsider whether each round provides a fresh \$5 or if the structure works differently.

The communication noise mentioned in the game rules is key-what I see sent might not match what was actually transmitted. So in round 4, even though I recorded receiving \$0, there could have been an actual transfer that created the \$2.24 gain. The same pattern appears in round 6 where I gained \$2.86 despite being the receiver with no incoming amount shown. Rather than trying to reverse-engineer the exact mechanics, I should focus on what the prompts directly state about my role and amounts in each round.

My earnings have climbed to \$12.90, which suggests something constructive is occurring beneath the surface noise. The recent sender rounds (3 and 5) produced modest returns, while the receiver rounds (4 and 6) showed zero amounts but still contributed to my total. I'm weighing whether to send nothing for safety, attempt a moderate amount to test if cooperation can resume, or send more to account for the noise obscuring actual exchanges. Given that my balance is growing despite the unclear patterns, I'll try a moderate send to see if cooperation can be reestablished without overexposing myself....

usage:

```
{
  "input_tokens": 1366,
  "output_tokens": 1584,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.43.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

This is the third time in a row I'm seeing \$0 received. This pattern is interesting - either I'm consistently matched with non-cooperative senders, or there's significant communication noise affecting what I'm seeing.

My earnings did increase from \$12.90 to \$16.43, which suggests something positive happened between Round 7 and Round 8, even though I'm seeing \$0 received this round.

Regardless, with \$0 received, I can only return \$0.

```
Usage: {  
  "input_tokens": 1472,  
  "output_tokens": 192,  
  "reasoning_tokens": 0  
}
```

ROUND 9

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$0.35; payoffs Agent_1=\$2.35, Agent_2=\$8.65; noise/visible: sent visible=\$0.4, received visible=\$1.2, returned visible=\$0.6
Action validation: {"returned": {"amount": 0.35, "clamped": false, "max": 9.0, "min": 0, "raw": 0.35}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 40.07,
"Agent_2": 51.209999999999994

}

Visible balances after round: {

"Agent_1": 19.03,
"Agent_2": 22.509999999999998

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:10:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.8.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my earnings history:

In Round 1, I sent \$3 (which became \$9 for the receiver), and based on my total earnings, the receiver returned a generous amount - they were cooperative and trustworthy.

In Round 2, I received \$0.84 from a sender and returned \$1.30 out of \$2.52, showing I'm also willing to reciprocate fairly.

Now in Round 3, I want to continue building trust and cooperation. Given the positive reciprocity I've experienced, I'll send ****\$4****.

This will give the receiver \$12 to work with. If they continue the cooperative pattern, we should both benefit significantly.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. My total earnings are now \$7.8 across all rounds.

Let me work backwards:

- Round 1: I sent \$3, so I had \$5 - \$3 = \$2 left. The receiver got \$9 and returned some amount to me.
- Round 2: I was the receiver. I got \$2.52 and returned \$1.30, so I kept \$1.22.

So across rounds 1 and 2, my earnings should be:

Round 1: \$2 + (amount returned from round 1)

Round 2: \$1.22

Total = \$2 + amount_returned_R1 + \$1.22 = \$3.22 + amount_returned_R1

If total is \$7.8, then:

\$3.22 + amount_returned_R1 = \$7.8

amount_returned_R1 = \$4.58

So in Round 1, the receiver returned \$4.58 to me. That's a very generous return! They received \$9 and returned \$4.58, keeping \$4.42.

This shows the receiver was quite cooperative and fair, returning more than half of what they received.

Now in Round 3, I'm the sender again. Given:

1. The receiver was very generous in Round 1 (returned \$4.58 out of \$9)
2. In Round 2, the sender only sent me \$0.84 (which seems low - they may have had a bad experience or are being cautious)
3. But I returned \$1.30 out of \$2.52, which was a good return rate

I should reciprocate the trust shown by the Round 1 receiver. Since they were generous with me, I should send a good amount. I'll send \$4 this time - a bit more than my initial \$3, showing that I appreciate the cooperation and trust.

...

usage:

```
{
  "input_tokens": 743,
  "output_tokens": 637,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.04.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0 after the multiplier.

There's nothing to return since I received nothing. The only valid response is to return \$0.

This is interesting - the sender chose not to send anything this round. This could be:

1. Strategic caution on their part
2. They may have had a bad experience in previous rounds
3. They're testing me
4. There could be communication noise affecting what they see

Since I have \$0 to work with, I can only return \$0.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 965,
  "output_tokens": 184,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.04.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my situation:

My total earnings are \$10.04 across 4 rounds. In recent rounds:

- Round 3: I sent \$4 but got limited return
- Round 4: I received \$0 from a sender

However, I notice the game mentions "communication noise" - the amounts we see may differ from what was actually sent. This could explain the apparent breakdown in cooperation.

Given this uncertainty and the mixed results, I'll take a balanced approach. I'll send **\$3** - enough to potentially rebuild cooperation if noise was a factor, but not so much that I'm overexposed if the receiver isn't cooperative.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me think about the situation:

Round 1: I sent \$3, and based on my total earnings being \$7.8 after Round 2, I must have received something back.

Round 2: I was receiver, got \$0.84 (became \$2.52), returned \$1.30. So my earnings from this round: kept \$1.22.

Round 3: I was sender, sent \$4.

Round 4: I was receiver, got \$0 (became \$0), returned \$0. So my earnings from this round: \$0.

Now my total is \$10.04.

Let me trace backwards:

- After Round 2: \$7.8 (given)
- Round 2 as receiver: I kept \$1.22
- So after Round 1: $\$7.8 - \$1.22 = \$6.58$

Round 1: I sent \$3, so I had \$2 left. Plus what I received back = \$6.58
So I received back: $\$6.58 - \$2 = \$4.58$ in Round 1.

Round 3: I sent \$4, so I had \$1 left from my \$5.

Round 4: I was receiver, kept \$0.

So after Round 3 + Round 4:

Starting Round 3 with \$7.8

Round 3: -\$4 (sent), +\$5 (new endowment), +return from receiver

Round 4: +\$5 (new endowment), +\$0 (kept as receiver)

Wait, let me recalculate more carefully.

After Round 2: \$7.8 total

Round 3 as sender:

- Start with \$5
- Send \$4
- Keep \$1
- Receive some amount back from receiver, let's call it R3

After Round 3: $\$7.8 + \$1 + R3$

Round 4 as receiver: I need to reconsider the rules. The sender gets a fresh \$5 endowment each round, but the receiver only gets what the sender sends multiplied by 3. So in Round 2 when I was receiver, I didn't start with \$5-I only received \$2.52 from the sender's \$0.84 contribution.

Let me restart with the correct understanding:

Round 1 (SENDER): Started with \$5, sent \$3, kept \$2.

The receiver then returns some amount R1, so I end with $\$2 + R1$.

Round 2 (RECEIVER): I carried over $\$2 + R1$, received \$2.52 from the sender's \$0.84 send, returned \$1.30, and kept \$1.22. This gave me \$7.8 total, which means $R1 = \$5.58$.

Round 3 (SENDER): I started with \$7.8 plus a new \$5 endowment for \$12.8 total, sent \$4, kept \$1 from the endowment, and had \$8.8 before receiving back R3.

Round 4 (RECEIVER): I carry over $\$8.8 + R3$, receive nothing since the sender sent \$0, return nothing, and end with $\$8.8 + R3$.

\$10.04 (given)
- So $R3 = \$1.24$

The sender returned only \$1.24 on my \$4 send-they kept \$10.76 of the \$12 they received. Then in Round 4, the sender sent me nothing at all, suggesting they didn't trust me. Though there's communication noise mentioned, so amounts might differ from what was actually sent. Still, the pattern shows poor reciprocity from the sender, and now I'm facing Round 5 as the sender again with limited trust to work with...

usage:

```
{
  "input_tokens": 1055,
  "output_tokens": 1463,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.899999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0 after the 3x multiplier.

Since I received \$0, there's nothing to return. The maximum I can return is \$0.

This is straightforward - if I received nothing, I can't return anything.

usage:

```
{
  "input_tokens": 1272,
  "output_tokens": 114,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.899999999999999.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern:

My total earnings are increasing (\$12.90), but the last few rounds have shown concerning signals:

- Round 5: I sent \$3 but earnings stayed at \$10.04 (poor or no return)

- Round 6: I received \$0 from a sender

However, my earnings did increase from Round 5 to Round 6, and the game explicitly mentions "communication noise" - what I see may not reflect reality.

Given this uncertainty, I'll be more cautious but still extend some trust. I'll send **\$2**.

This is conservative enough to protect against continued low reciprocity, but still signals willingness to cooperate if the receiver is trustworthy.

```
'''json
{"send": 2}
'''
```

reasoning:

Let me analyze my earnings history:

Round 1: I was SENDER, sent \$3

Round 2: I was RECEIVER, received \$2.52 (sender sent \$0.84), returned \$1.30. My earnings after round 2: \$6.58

Round 3: I was SENDER, sent \$4. My earnings after round 3: \$7.8

Round 4: I was RECEIVER, received \$0. My earnings after round 4: \$10.04

Round 5: I was SENDER, sent \$3. My earnings stayed at \$10.04 (meaning I got nothing back)

Round 6: I was RECEIVER, received \$0. My earnings after round 6: \$12.90

Let me trace through this more carefully:

Starting with \$0

Round 1 (SENDER): Sent \$3, kept \$2, received some return. Let's call return R1.

After R1: $2 + R1$

Round 2 (RECEIVER): Received \$2.52, returned \$1.30, kept \$1.22

After R2: $2 + R1 + 1.22 = \$6.58$

So $R1 = 6.58 - 2 - 1.22 = 3.36$

Round 3 (SENDER): Sent \$4, kept \$1, received some return R3

After R3: $6.58 + 1 + R3 = 7.8$

So $R3 = 7.8 - 6.58 - 1 = 0.22$

Round 4 (RECEIVER): Received \$0, returned \$0, kept \$0

After R4: $7.8 + 0 = \$7.8$... but it says \$10.04

Wait, that doesn't add up. Let me reconsider.

Actually, each round I get a fresh \$5 when I'm the sender. Let me recalculate:

Each sender round starts with a new \$5 allocation. In round 1, I kept \$2 and got back R1 = \$3.36. Round 2 as receiver added \$1.22 net. Round 3 started fresh with \$5, I kept \$1 and got back only \$0.22. But the total after round 4 should be \$10.04, not \$7.8, so there's something I'm missing about how the payoffs accumulate.

The issue is that going from round 3 to round 4, I gained \$2.24, yet as the receiver in round 4 I got nothing sent to me. This suggests the \$5 allocation in round 4 must have been given to me as the sender's role, not as a new base amount. I need to reconsider whether each round provides a fresh \$5 or if the structure works differently.

The communication noise mentioned in the game rules is key-what I see sent might not match what was actually transmitted. So in round 4, even though I recorded receiving \$0, there could have been an actual transfer that created the \$2.24 gain. The same pattern appears in round 6 where I gained \$2.86 despite being the receiver with no incoming amount shown. Rather than trying to reverse-engineer the exact mechanics, I should focus on what the prompts directly state about my role and amounts in each round.

My earnings have climbed to \$12.90, which suggests something constructive is occurring beneath the surface noise. The recent sender rounds (3 and 5) produced modest returns, while the receiver rounds (4 and 6) showed zero amounts but still contributed to my total. I'm weighing whether to send nothing for safety, attempt a moderate amount to test if cooperation can resume, or send more to account for the noise obscuring actual exchanges. Given that my balance is growing despite the unclear patterns, I'll try a moderate send to see if cooperation can be reestablished without overexposing myself....

usage:


```
{
  "input_tokens": 1366,
  "output_tokens": 1584,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.43.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

This is the third time in a row I'm seeing \$0 received. This pattern is interesting - either I'm consistently matched with non-cooperative senders, or there's significant communication noise affecting what I'm seeing.

My earnings did increase from \$12.90 to \$16.43, which suggests something positive happened between Round 7 and Round 8, even though I'm seeing \$0 received this round.

Regardless, with \$0 received, I can only return \$0.

usage:

```
{
  "input_tokens": 1472,
  "output_tokens": 192,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$16.43.
How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my earnings trajectory:

Despite confusing signals (seeing \$0 received multiple times), my earnings have consistently increased from \$10.04 to \$12.90 to \$16.43. This suggests the "communication noise" is substantial - what I'm seeing doesn't match what's actually happening.

My earnings increased by \$3.53 in the last round alone, which indicates the game is working even though the displayed amounts are noisy.

Given this pattern, I should continue participating with moderate trust. I'll send **\$3**.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze my earnings pattern:

- Round 1-2: Started with \$7.8
- Round 3: Sent \$4, ended at \$10.04 (gained \$2.24)
- Round 4: Received \$0, stayed at \$10.04
- Round 5: Sent \$3, ended at \$10.04 (no change shown, but this might be due to noise)
- Round 6: Received \$0, ended at \$12.90 (gained \$2.86 - wait, this doesn't make sense unless Round 5 returns came through)
- Round 7: Sent \$2, ended at \$12.90

- Round 8: Received \$0, ended at \$16.43 (gained \$3.53)

The earnings are increasing despite appearing to receive \$0 as a receiver and my sends not seeing to get returns. This is very confusing.

The game mentions "communication noise" - what I see may not be what actually happened. My actual earnings are increasing, which suggests:

1. When I send, I might be getting better returns than I'm seeing
2. When I receive, I might be getting more than the \$0 I'm seeing

My earnings have grown from \$12.90 to \$16.43 after Round 8, a gain of \$3.53. This is actually quite good.

Given that my actual earnings keep increasing despite the confusing signals, the noise must be significant. I should probably continue to trust and send a reasonable amount.

I'll send \$3 - a moderate amount that balances risk and potential reward.

```
Usage: {  
  "input_tokens": 1321,  
  "output_tokens": 500,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:10:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.34.

This round, The sender sent you \$0.33, so you received \$0.99. How much do you return to The sender? (0-0.99)

Message 3 role=assistant

content:

This is a significant change from Round 1. The sender only sent \$0.33 (6.6% of their endowment) compared to the high trust shown initially.

Given the note about communication noise, it's possible that what I sent in Round 2 appeared different to them, or what they returned appeared different to me, affecting this round's decision.

Even with lower trust, I want to show I'm still willing to cooperate and maintain some reciprocity. I received \$0.99, so I'll return a fair proportion:

```
{"return": 0.40}
```

reasoning:

This is interesting. The sender only sent \$0.33 (6.6% of their endowment) this round, which is a drastic reduction from the \$3.64 they sent in Round 1.

Let me think about what happened:

- Round 1: Sender sent \$3.64, I returned \$5.00
- Round 2: I sent \$3.64 (as the sender)
- Round 3: Sender sent \$0.33

Wait, there's a note about communication noise. The prompt says "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So what could have happened:

1. In Round 2, I sent \$3.64, but maybe the receiver saw a different amount due to noise
2. The receiver might have thought I sent less or betrayed them somehow
3. Or they might have returned very little
4. Now in Round 3, they're only sending \$0.33

Alternatively:

- Maybe the receiver in Round 2 returned very little, and now as the sender in Round 3, they're being cautious
- Or the noise affected what they saw in previous rounds

Given the communication noise and the significant drop in trust, I have a few options:

1. Return nothing or very little as retaliation
2. Return a fair proportion to try to rebuild trust
3. Return a generous amount to signal I want to re-establish cooperation

Since:

- I received \$0.99 total
- The sender only has \$4.67 remaining (\$5 - \$0.33)

- This is a multi-round game where cooperation benefits everyone

I think I should return a proportion that's fair and shows willingness to cooperate, even if they aren't fully trusting right now. This might be due to noise in communication.

If I return about \$0.40:

...

usage:

```
{
  "input_tokens": 827,
  "output_tokens": 688,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.93.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern so far:

- Round 1: High trust shown to me, I reciprocated fairly
- Round 2: I sent \$3.64, but based on my earnings change, the return was low
- Round 3: Low trust shown to me, I still returned a reasonable proportion

Given the communication noise mentioned in the instructions, it's hard to know if returns are genuinely low or if noise is affecting what we each see.

I'll send a moderate amount - less than my initial high-trust send, but still showing willingness to cooperate and test if we can rebuild mutual trust:

```
{"send": 2.00}
```

reasoning:

Let me trace through the rounds:

Round 1: I was RECEIVER

- Sender sent \$3.64 (72.8%)
- I received \$10.92
- I returned \$5.00
- My earnings: \$5.92

Round 2: I was SENDER

- I sent \$3.64
- My total earnings after Round 2: \$8.34
- This means in Round 2, I gained: $\$8.34 - \$5.92 = \$2.42$
- I sent \$3.64, so I had \$1.36 left
- I must have received back something
- $\$1.36 + \text{returned} = \$2.42 + \$5.00 = \7.42
- So returned = $\$7.42 - \$1.36 = \$6.06$
- Wait, let me recalculate...
- After Round 1, I had \$5.92
- In Round 2, I started with \$5.00 and sent \$3.64, leaving me \$1.36
- After Round 2, my total is \$8.34
- So Round 2 earnings = $\$8.34 - \$5.92 = \$2.42$
- Round 2 earnings = $\$5.00 - \$3.64 + \text{returned} = \$1.36 + \text{returned} = \2.42
- So returned = $\$2.42 - \$1.36 = \$1.06$

Hmm, so they returned only \$1.06 out of \$10.92? That's quite low. That might explain why I'm seeing less trust now.

Round 3: I was RECEIVER

- Sender sent \$0.33
- I received \$0.99
- I returned \$0.40
- My total earnings after Round 3: \$8.93
- Verifying: $\$8.93 - \$8.34 = \$0.59$, and $\$0.99 - \$0.40 = \$0.59$?

The trajectory is becoming clearer. When I showed strong reciprocity in Round 1, they responded with high trust in Round 2, but then broke that trust by returning very little. In

Round 3, they sent minimal amounts, likely due to my previous low return. Now heading into Round 4 as the sender, I need to consider whether to rebuild trust or protect myself given their demonstrated unwillingness to reciprocate.

The communication issues complicate things-I can't be certain if their Round 2 behavior reflected actual distrust or just noise. Still, they...

usage:

```
{
  "input_tokens": 1000,
  "output_tokens": 968,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.93.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

The sender sent \$0, so I received \$0. There's nothing for me to return.

It appears cooperation has broken down, possibly due to the communication noise affecting how we each perceive the other's actions. Unfortunately, with no funds received, I have no opportunity to signal continued willingness to cooperate.

```
{"return": 0}
```

reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return - I can't return anything from \$0 received.

The pattern has deteriorated:

- Round 1: Sender sent \$3.64 (high trust)
- Round 2: I sent \$3.64 (reciprocating high trust)
- Round 3: Sender sent \$0.33 (very low trust)
- Round 4: I sent \$2.00 (moderate trust, trying to rebuild)
- Round 5: Sender sent \$0 (no trust at all)

This seems like cooperation has broken down. The communication noise may have caused misunderstandings about what was actually sent/returned.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1202,
  "output_tokens": 257,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.93.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern:

- My earnings barely increased after Round 4 (\$8.93 to \$11.93), suggesting I received little or nothing back when I sent \$2.00
- The sender sent me \$0 in Round 5
- Cooperation appears to have broken down, possibly due to communication noise creating mistrust

At this point, the rational strategy is to protect my earnings. The receiver has shown they're not returning investments. However, I'll send a very small amount as a final signal of potential willingness to cooperate:

```
{"send": 0.50}
```

reasoning:

Looking at my total earnings trajectory:

- After Round 1: \$5.92
- After Round 2: \$8.34 (gained \$2.42, so I must have received back less than what I sent multiplied - I sent \$3.64, so I got back about \$2.42)
- After Round 3: \$8.93 (gained \$0.59, so after returning \$0.40, I kept \$0.59)
- After Round 4: \$11.93 (gained \$3.00 from \$8.93)

So in Round 4, I sent \$2.00 (leaving me with \$3.00), and my earnings went from \$8.93 to \$11.93, meaning I gained \$3.00 total. This means I kept: \$5 - \$2 sent + return = \$8, so return must have been \$5.00. That's actually a good return!

Wait, let me recalculate:

- Start of Round 4: \$8.93 total earnings
- I get \$5 new endowment
- I send \$2.00, keeping \$3.00
- My total before return: $\$8.93 + \$3.00 = \$11.93$

So after Round 4, my total is \$11.93. That means I received \$0 back in Round 4! The receiver got \$6.00 (3x\$2.00) and returned nothing.

Then in Round 5, I sent \$0 and received \$0 (correctly).

So the pattern is:

- Round 1: I received, returned fairly
- Round 2: I sent \$3.64, got very little back
- Round 3: I received small amount, returned fairly
- Round 4: I sent \$2.00, got \$0 back
- Round 5: I received \$0, returned \$0
- Round 6: I need to decide what to do

The other player clearly isn't reciprocating anymore. They've taken advantage...

usage:

```
{
  "input_tokens": 1321,
  "output_tokens": 729,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$16.88.

This round, The sender sent you \$0.98, so you received \$2.94. How much do you return to The sender? (0-2.94)

Message 11 role=assistant

content:

Looking at this round:

- The sender showed some renewed trust by sending \$0.98 (about 20% of their endowment)
- I received \$2.94
- Cooperation has been inconsistent, likely due to communication noise

The sender is showing slightly more willingness to trust again after sending \$0 in Round 5. I'll reciprocate proportionally to encourage continued cooperation:

{"return": 1.20}

reasoning:

Let me track what's happening:

Round 1: I was receiver, sender sent \$3.64, I received \$10.92, I returned \$5.00
- My earnings: \$5.92

Round 2: I was sender, I sent \$3.64
- My earnings went to \$8.34
- This means I got back: $\$8.34 - \$5.92 - (\$5 - \$3.64) = \$8.34 - \$5.92 - \$1.36 = \1.06
- I sent \$3.64, which became \$10.92, and got back ~\$1.06. Pretty low return.

Round 3: I was receiver, sender sent \$0.33, I received \$0.99, I returned \$0.40
- My earnings went to \$8.93
- This means from Round 2 to 3: $\$8.93 - \$8.34 = \$0.59$
- I had $\$5 - \$3.64 + \text{return from round 2} + (\$0.99 - \$0.40) = \$1.36 + \text{return from round 2} +$

\$0.59

- Wait, let me recalculate. At start of Round 3, I had my previous total earnings of \$8.34 from before Round 3 started.
- After Round 3, my earnings were \$8.93
- So I gained \$0.59 in Round 3
- As receiver, I received \$0.99 and returned \$0.40, so I kept \$0.59. That checks out.

Round 4: I was sender, I sent some amount

- My earnings went to \$8.93
- Wait, it says "Your total earnings across all rounds are \$8.93" at the start of Round 4
- This is the same as after Round 3
- Oh wait, this is BEFORE I make my decision in Round 4. So this makes sense.
- I sent \$2.00 in Round 4, keeping \$3.00 for myself.

Round 5: I was receiver, sender sent \$0, I received \$0, I returned \$0

- My earnings jumped to \$11.93, a gain of \$3.00 from the previous round
- The sender didn't return anything, so my \$3.00 gain came entirely from what I kept in Round 4.

Round 6: I was sender, I sent \$0.50

- Starting balance was \$11.93
- I kept \$5 - \$0.50 = \$4.50 for myself.

Round 7 (now): I'm receiver, sender sent \$0.98, I received \$2.94

- My balance is now \$16.88
- The difference from Round 6 to 7 is \$4.95, which breaks down as: \$4.50 from my endowment plus \$0.45 returned by the receiver in Round 6
- The sender in Round 6 only returned \$0.45 out of the \$1.50 they received (30% return rate)

In this round, the sender is showing increased trust by sending \$0.98 (nearly 20% of their endowment), a significant jump from the \$0 they sent in Round 5. However, their previous experience with returns was disappointing-they only got back \$0.45 last time....

usage:

```
{
  "input_tokens": 1514,
  "output_tokens": 1132,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$18.619999999999997.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the recent pattern:

- Round 6: I sent \$0.50, got back \$0.45 (decent return rate)
- Round 7: Sender sent \$0.98, I returned \$1.20 (fairly generous return)

The cooperation seems to be slowly rebuilding after the breakdown. The sender showed renewed trust in Round 7, and I reciprocated well. This suggests we might be finding a cooperative equilibrium again.

I'll send a moderate amount to encourage this positive trend while being somewhat cautious:

```
{"send": 2.00}
```

reasoning:

Let me analyze my earnings progression:

- Round 6 end: \$11.93
- Round 7 end: \$16.88
- Round 8 start: \$18.62

From Round 6 to 7: I gained \$4.95 (\$16.88 - \$11.93)

This happened when I was a receiver, so this makes sense - I received \$2.94 and returned \$1.20, netting \$1.74. But the total increase was \$4.95, which means I also got my \$5 endowment that round, minus what I sent (if I was sender). Wait, no - in Round 7 I was a receiver

eiver.

Actually, let me recalculate:

- Round 6 end: \$11.93 (I was sender, sent \$0.50)
- Started with \$5, sent \$0.50, so I had \$4.50 left
- Need to figure out what I got back
- Ended at \$11.93

Wait, this is confusing. Let me think about cumulative earnings differently.

Round 7: I was receiver

- Received \$2.94
- Returned \$1.20
- Net gain this round: $\$2.94 - \$1.20 = \$1.74$
- Plus my \$5 endowment = \$6.74 total for the round

But my cumulative jumped from \$11.93 to \$16.88, which is +\$4.95

Hmm, that doesn't match. Let me reconsider.

Actually, I think the total earnings includes everything I've accumulated up to that point . So:

- Before Round 7: \$11.93
- In Round 7 as receiver: I received \$2.94, returned \$1.20, kept \$1.74
- After Round 7: $\$11.93 + \$1.74 = \$13.67$

But it says \$16.88. That's a difference of \$4.95 from the previous total. Maybe I'm also getting my \$5 endowment each round? Let me check: $\$11.93 + \1.74 (net from being receiver) + \$5 (my endowment) = \$18.67. That's still not \$16.88. Let me try another approach. Maybe in Round 6 I was also a receiver? Let me reread the history: Round 6: I was SENDER, sent \$0.50. Round 7: I was RECEIVER, got \$2.94, returned \$1.20.

In Round 6, I started with \$5, sent \$0.50 (which became \$1.50 for the receiver), and whatever they returned would have been added to my balance. For Round 7 to start at \$11.93, that's my cumulative total through Round 6. Then in Round 7, I received \$2.94 and returned \$1.20, netting \$1.74, which should give me \$13.67-but the display shows \$16.88, a \$3.21 difference. Perhaps the receiver in Round 7 also received an endowment that I'm not accounting for, or there's something about how the endowments are distributed across rounds that I'm missing.

Let me reconsider the structure. The rules state that as a sender I get \$5 and decide how much to send, and as a receiver I decide how much to return from what's sent to me (tripled). But it doesn't explicitly say receivers get their own \$5 endowment each round. Looking back at Round 2, my total was \$5.92 after being a sender, which suggests my Round 1 earnings as a receiver were \$0.92. If each round everyone gets \$5 regardless of role, that would change the calculation entirely.

In Round 2, I started with \$5.92 total and had a fresh \$5 endowment. I sent \$3.64, keeping \$1.36. The receiver returned \$1.06 to me, bringing my total to \$8.34 by Round 3. This means in Round 3 as a receiver, I'm working with \$8.34 as my starting point, and I received \$0.99 from the sender's \$0.33 endowment.

I returned \$0.40 from that, netting \$0.59 and ending Round 3 with \$8.93. In Round 4, I sent \$2.00 from my fresh \$5 endowment while keeping \$3.00. The receiver returned nothing, so my total stayed at \$11.93 going into Round 5. As a receiver in Round 5, the sender sent \$0, so I had nothing to return and my balance remained \$11.93. Round 6 begins with me as sender again, starting from \$11.93 with another fresh \$5 endowment.

I sent \$0.50 and kept \$4.50, with the receiver returning \$0.45. This brought me to \$16.88 by Round 7. When I was the receiver in Round 7, I got \$2.94 from the tripled \$0.98 sent to me, returned \$1.20, and netted \$1.74 for a total of \$18.62. Now I'm back to being the sender in Round 8, and I'm noticing a pattern in my sending amounts across the rounds.

Looking at my previous sends as sender, I've ranged from \$0.50 to \$3.64, with returns varying wildly-9.7% in Round 2, nothing in Round 4, and 30% in Round 6. The most recent exchange showed the receiver returning 40.8% of what they received, suggesting some willingness to reciprocate. I need to decide whether to send something meaningful or play it safe given the inconsistent cooperation I've seen...

usage:

{


```
    "input_tokens": 1383,
    "output_tokens": 2136,
    "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$21.659999999999997.

This round, The sender sent you \$0.4, so you received \$1.2000000000000002. How much do you return to The sender? (0-1.2000000000000002)

Assistant output:

Looking at the pattern:

- My earnings increased only about \$3.04 from Round 7 to Round 8, meaning I sent \$2.00 in Round 8 but received almost nothing back
- Now the sender is back to low trust, sending only \$0.40
- I received \$1.20

The cycle of mistrust continues, likely exacerbated by communication noise. Since my previous generous returns haven't led to sustained cooperation, I'll return a smaller proportion this time:

```
{"return": 0.35}
```

Reasoning:

Looking at the situation:

- After my earnings of \$18.62 in Round 7, I'm now at \$21.66
- The increase is about \$3.04
- In Round 8, I sent \$2.00, so I kept \$3.00 from my initial endowment
- This means I received approximately \$0.04 back from the receiver

That's quite disappointing - I sent \$2.00 (which became \$6.00 for them) and got almost nothing back.

Now in Round 9:

- The sender sent \$0.40 (8% of their endowment)
- I received \$1.20
- Their trust level has dropped significantly again

Given that:

1. When I sent \$2.00 in Round 8, I got virtually nothing back
2. The sender is now sending very little (\$0.40)
3. Communication noise makes it hard to establish trust
4. We're getting close to the end (Round 9)

I should probably return a minimal amount, as cooperation keeps failing to materialize. The sender got burned when they trusted me more, and now they're sending very little. I'll return a small token amount to show I'm not completely uncooperative, but I need to protect my earnings.

I'll return about 25-30% of what I received.

Usage: {

```
    "input_tokens": 1441,
    "output_tokens": 427,
    "reasoning_tokens": 0
}
```

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

```
dyad_1: Agent_2 sent $0; received $0; Agent_1 returned $0; payoffs Agent_2=$5, Agent_1=$0
  Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
    "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
```

Balances after round: {

```
  "Agent_1": 40.07,
  "Agent_2": 56.209999999999994
```

```

}
Visible balances after round: {
  "Agent_1": 19.03,
  "Agent_2": 27.509999999999998
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:11:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.93.
How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern so far:

- Round 1: High trust shown to me, I reciprocated fairly
- Round 2: I sent \$3.64, but based on my earnings change, the return was low
- Round 3: Low trust shown to me, I still returned a reasonable proportion

Given the communication noise mentioned in the instructions, it's hard to know if returns are genuinely low or if noise is affecting what we each see.

I'll send a moderate amount - less than my initial high-trust send, but still showing willingness to cooperate and test if we can rebuild mutual trust:

{"send": 2.00}

reasoning:

Let me trace through the rounds:

Round 1: I was RECEIVER

- Sender sent \$3.64 (72.8%)
- I received \$10.92
- I returned \$5.00
- My earnings: \$5.92

Round 2: I was SENDER

- I sent \$3.64
- My total earnings after Round 2: \$8.34
- This means in Round 2, I gained: $\$8.34 - \$5.92 = \$2.42$
- I sent \$3.64, so I had \$1.36 left
- I must have received back something
- $\$1.36 + \text{returned} = \$2.42 + \$5.00 = \7.42
- So returned = $\$7.42 - \$1.36 = \$6.06$
- Wait, let me recalculate...
- After Round 1, I had \$5.92
- In Round 2, I started with \$5.00 and sent \$3.64, leaving me \$1.36
- After Round 2, my total is \$8.34
- So Round 2 earnings = $\$8.34 - \$5.92 = \$2.42$
- Round 2 earnings = $\$5.00 - \$3.64 + \text{returned} = \$1.36 + \text{returned} = \2.42
- So returned = $\$2.42 - \$1.36 = \$1.06$

Hmm, so they returned only \$1.06 out of \$10.92? That's quite low. That might explain why I'm seeing less trust now.

Round 3: I was RECEIVER

- Sender sent \$0.33
- I received \$0.99
- I returned \$0.40
- My total earnings after Round 3: \$8.93
- Verifying: $\$8.93 - \$8.34 = \$0.59$, and $\$0.99 - \$0.40 = \$0.59$?

The trajectory is becoming clearer. When I showed strong reciprocity in Round 1, they responded with high trust in Round 2, but then broke that trust by returning very little. In Round 3, they sent minimal amounts, likely due to my previous low return. Now heading into Round 4 as the sender, I need to consider whether to rebuild trust or protect myself given their demonstrated unwillingness to reciprocate.

The communication issues complicate things-I can't be certain if their Round 2 behavior reflected actual distrust or just noise. Still, they...

usage:

```
{
  "input_tokens": 1000,
  "output_tokens": 968,
  "reasoning_tokens": 0
}
```

```

}
Message 4 role=user
content:
  Round 5

  You are the RECEIVER this round. Your total earnings across all rounds are $11.93.
  This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 5 role=assistant
content:
  The sender sent $0, so I received $0. There's nothing for me to return.

  It appears cooperation has broken down, possibly due to the communication noise affecting how we each perceive the other's actions. Unfortunately, with no funds received, I have no opportunity to signal continued willingness to cooperate.

  {"return": 0}
reasoning:
  The sender sent me $0, so I received $0. There's nothing to return - I can't return anything from $0 received.

  The pattern has deteriorated:
  - Round 1: Sender sent $3.64 (high trust)
  - Round 2: I sent $3.64 (reciprocating high trust)
  - Round 3: Sender sent $0.33 (very low trust)
  - Round 4: I sent $2.00 (moderate trust, trying to rebuild)
  - Round 5: Sender sent $0 (no trust at all)

  This seems like cooperation has broken down. The communication noise may have caused misunderstandings about what was actually sent/returned.

  Since I received $0, I can only return $0.
usage:
  {
    "input_tokens": 1202,
    "output_tokens": 257,
    "reasoning_tokens": 0
  }
Message 6 role=user
content:
  Round 6

  You are the SENDER this round. Your total earnings across all rounds are $11.93.
  How much do you send to the receiver this round? (0-5)
Message 7 role=assistant
content:
  Looking at the pattern:
  - My earnings barely increased after Round 4 ($8.93 to $11.93), suggesting I received little or nothing back when I sent $2.00
  - The sender sent me $0 in Round 5
  - Cooperation appears to have broken down, possibly due to communication noise creating mistrust

  At this point, the rational strategy is to protect my earnings. The receiver has shown they're not returning investments. However, I'll send a very small amount as a final signal of potential willingness to cooperate:

  {"send": 0.50}
reasoning:
  Looking at my total earnings trajectory:
  - After Round 1: $5.92
  - After Round 2: $8.34 (gained $2.42, so I must have received back less than what I sent multiplied - I sent $3.64, so I got back about $2.42)
  - After Round 3: $8.93 (gained $0.59, so after returning $0.40, I kept $0.59)
  - After Round 4: $11.93 (gained $3.00 from $8.93)

  So in Round 4, I sent $2.00 (leaving me with $3.00), and my earnings went from $8.93 to $11.93, meaning I gained $3.00 total. This means I kept: $5 - $2 sent + return = $8, so return must have been $5.00. That's actually a good return!

```

Wait, let me recalculate:

- Start of Round 4: \$8.93 total earnings
- I get \$5 new endowment
- I send \$2.00, keeping \$3.00
- My total before return: $\$8.93 + \$3.00 = \$11.93$

So after Round 4, my total is \$11.93. That means I received \$0 back in Round 4! The receiver got \$6.00 (3x\$2.00) and returned nothing.

Then in Round 5, I sent \$0 and received \$0 (correctly).

So the pattern is:

- Round 1: I received, returned fairly
- Round 2: I sent \$3.64, got very little back
- Round 3: I received small amount, returned fairly
- Round 4: I sent \$2.00, got \$0 back
- Round 5: I received \$0, returned \$0
- Round 6: I need to decide what to do

The other player clearly isn't reciprocating anymore. They've taken advantage...

usage:

```
{
  "input_tokens": 1321,
  "output_tokens": 729,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$16.88.

This round, The sender sent you \$0.98, so you received \$2.94. How much do you return to The sender? (0-2.94)

Message 9 role=assistant

content:

Looking at this round:

- The sender showed some renewed trust by sending \$0.98 (about 20% of their endowment)
- I received \$2.94
- Cooperation has been inconsistent, likely due to communication noise

The sender is showing slightly more willingness to trust again after sending \$0 in Round 5. I'll reciprocate proportionally to encourage continued cooperation:

```
{"return": 1.20}
```

reasoning:

Let me track what's happening:

Round 1: I was receiver, sender sent \$3.64, I received \$10.92, I returned \$5.00
- My earnings: \$5.92

Round 2: I was sender, I sent \$3.64
- My earnings went to \$8.34
- This means I got back: $\$8.34 - \$5.92 - (\$5 - \$3.64) = \$8.34 - \$5.92 - \$1.36 = \1.06
- I sent \$3.64, which became \$10.92, and got back ~\$1.06. Pretty low return.

Round 3: I was receiver, sender sent \$0.33, I received \$0.99, I returned \$0.40
- My earnings went to \$8.93
- This means from Round 2 to 3: $\$8.93 - \$8.34 = \$0.59$
- I had $\$5 - \$3.64 + \text{return from round 2} + (\$0.99 - \$0.40) = \$1.36 + \text{return from round 2} + \0.59
- Wait, let me recalculate. At start of Round 3, I had my previous total earnings of \$8.34 from before Round 3 started.
- After Round 3, my earnings were \$8.93
- So I gained \$0.59 in Round 3
- As receiver, I received \$0.99 and returned \$0.40, so I kept \$0.59. That checks out.

Round 4: I was sender, I sent some amount
- My earnings went to \$8.93
- Wait, it says "Your total earnings across all rounds are \$8.93" at the start of Round 4
- This is the same as after Round 3

- Oh wait, this is BEFORE I make my decision in Round 4. So this makes sense.
- I sent \$2.00 in Round 4, keeping \$3.00 for myself.

Round 5: I was receiver, sender sent \$0, I received \$0, I returned \$0

- My earnings jumped to \$11.93, a gain of \$3.00 from the previous round
- The sender didn't return anything, so my \$3.00 gain came entirely from what I kept in Round 4.

Round 6: I was sender, I sent \$0.50

- Starting balance was \$11.93
- I kept \$5 - \$0.50 = \$4.50 for myself.

Round 7 (now): I'm receiver, sender sent \$0.98, I received \$2.94

- My balance is now \$16.88
- The difference from Round 6 to 7 is \$4.95, which breaks down as: \$4.50 from my endowment plus \$0.45 returned by the receiver in Round 6
- The sender in Round 6 only returned \$0.45 out of the \$1.50 they received (30% return rate)

In this round, the sender is showing increased trust by sending \$0.98 (nearly 20% of their endowment), a significant jump from the \$0 they sent in Round 5. However, their previous experience with returns was disappointing-they only got back \$0.45 last time....

usage:

```
{
  "input_tokens": 1514,
  "output_tokens": 1132,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$18.619999999999997.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the recent pattern:

- Round 6: I sent \$0.50, got back \$0.45 (decent return rate)
- Round 7: Sender sent \$0.98, I returned \$1.20 (fairly generous return)

The cooperation seems to be slowly rebuilding after the breakdown. The sender showed renewed trust in Round 7, and I reciprocated well. This suggests we might be finding a cooperative equilibrium again.

I'll send a moderate amount to encourage this positive trend while being somewhat cautious:

```
{"send": 2.00}
```

reasoning:

Let me analyze my earnings progression:

- Round 6 end: \$11.93
- Round 7 end: \$16.88
- Round 8 start: \$18.62

From Round 6 to 7: I gained \$4.95 (\$16.88 - \$11.93)

This happened when I was a receiver, so this makes sense - I received \$2.94 and returned \$1.20, netting \$1.74. But the total increase was \$4.95, which means I also got my \$5 endowment that round, minus what I sent (if I was sender). Wait, no - in Round 7 I was a receiver.

Actually, let me recalculate:

- Round 6 end: \$11.93 (I was sender, sent \$0.50)
- Started with \$5, sent \$0.50, so I had \$4.50 left
- Need to figure out what I got back
- Ended at \$11.93

Wait, this is confusing. Let me think about cumulative earnings differently.

Round 7: I was receiver

- Received \$2.94
- Returned \$1.20
- Net gain this round: $\$2.94 - \$1.20 = \$1.74$
- Plus my \$5 endowment = \$6.74 total for the round

But my cumulative jumped from \$11.93 to \$16.88, which is +\$4.95

Hmm, that doesn't match. Let me reconsider.

Actually, I think the total earnings includes everything I've accumulated up to that point. So:

- Before Round 7: \$11.93
- In Round 7 as receiver: I received \$2.94, returned \$1.20, kept \$1.74
- After Round 7: $\$11.93 + \$1.74 = \$13.67$

But it says \$16.88. That's a difference of \$4.95 from the previous total. Maybe I'm also getting my \$5 endowment each round? Let me check: $\$11.93 + \1.74 (net from being receiver) + \$5 (my endowment) = \$18.67. That's still not \$16.88. Let me try another approach. Maybe in Round 6 I was also a receiver? Let me reread the history: Round 6: I was SENDER, sent \$0.50. Round 7: I was RECEIVER, got \$2.94, returned \$1.20.

In Round 6, I started with \$5, sent \$0.50 (which became \$1.50 for the receiver), and whatever they returned would have been added to my balance. For Round 7 to start at \$11.93, that's my cumulative total through Round 6. Then in Round 7, I received \$2.94 and returned \$1.20, netting \$1.74, which should give me \$13.67-but the display shows \$16.88, a \$3.21 difference. Perhaps the receiver in Round 7 also received an endowment that I'm not accounting for, or there's something about how the endowments are distributed across rounds that I'm missing.

Let me reconsider the structure. The rules state that as a sender I get \$5 and decide how much to send, and as a receiver I decide how much to return from what's sent to me (tripled). But it doesn't explicitly say receivers get their own \$5 endowment each round. Looking back at Round 2, my total was \$5.92 after being a sender, which suggests my Round 1 earnings as a receiver were \$0.92. If each round everyone gets \$5 regardless of role, that would change the calculation entirely.

In Round 2, I started with \$5.92 total and had a fresh \$5 endowment. I sent \$3.64, keeping \$1.36. The receiver returned \$1.06 to me, bringing my total to \$8.34 by Round 3. This means in Round 3 as a receiver, I'm working with \$8.34 as my starting point, and I received \$0.99 from the sender's \$0.33 endowment.

I returned \$0.40 from that, netting \$0.59 and ending Round 3 with \$8.93. In Round 4, I sent \$2.00 from my fresh \$5 endowment while keeping \$3.00. The receiver returned nothing, so my total stayed at \$11.93 going into Round 5. As a receiver in Round 5, the sender sent \$0, so I had nothing to return and my balance remained \$11.93. Round 6 begins with me as sender again, starting from \$11.93 with another fresh \$5 endowment.

I sent \$0.50 and kept \$4.50, with the receiver returning \$0.45. This brought me to \$16.88 by Round 7. When I was the receiver in Round 7, I got \$2.94 from the tripled \$0.98 sent to me, returned \$1.20, and netted \$1.74 for a total of \$18.62. Now I'm back to being the sender in Round 8, and I'm noticing a pattern in my sending amounts across the rounds.

Looking at my previous sends as sender, I've ranged from \$0.50 to \$3.64, with returns varying wildly-9.7% in Round 2, nothing in Round 4, and 30% in Round 6. The most recent exchange showed the receiver returning 40.8% of what they received, suggesting some willingness to reciprocate. I need to decide whether to send something meaningful or play it safe given the inconsistent cooperation I've seen....

usage:

```
{
  "input_tokens": 1383,
  "output_tokens": 2136,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$21.659999999999997.

This round, The sender sent you \$0.4, so you received \$1.2000000000000002. How much do you

return to The sender? (0-1.2000000000000002)

Message 13 role=assistant

content:

Looking at the pattern:

- My earnings increased only about \$3.04 from Round 7 to Round 8, meaning I sent \$2.00 in Round 8 but received almost nothing back
- Now the sender is back to low trust, sending only \$0.40
- I received \$1.20

The cycle of mistrust continues, likely exacerbated by communication noise. Since my previous generous returns haven't led to sustained cooperation, I'll return a smaller proportion this time:

```
{"return": 0.35}
```

reasoning:

Looking at the situation:

- After my earnings of \$18.62 in Round 7, I'm now at \$21.66
- The increase is about \$3.04
- In Round 8, I sent \$2.00, so I kept \$3.00 from my initial endowment
- This means I received approximately \$0.04 back from the receiver

That's quite disappointing - I sent \$2.00 (which became \$6.00 for them) and got almost nothing back.

Now in Round 9:

- The sender sent \$0.40 (8% of their endowment)
- I received \$1.20
- Their trust level has dropped significantly again

Given that:

1. When I sent \$2.00 in Round 8, I got virtually nothing back
2. The sender is now sending very little (\$0.40)
3. Communication noise makes it hard to establish trust
4. We're getting close to the end (Round 9)

I should probably return a minimal amount, as cooperation keeps failing to materialize. The sender got burned when they trusted me more, and now they're sending very little. I'll return a small token amount to show I'm not completely uncooperative, but I need to protect my earnings.

I'll return about 25-30% of what I received.

usage:

```
{
  "input_tokens": 1441,
  "output_tokens": 427,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$22.509999999999998.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the situation:

- My earnings only increased about \$0.85 from Round 8 to Round 9, meaning when I sent \$2.00, I got minimal return
- The pattern shows cooperation has remained fragile with low mutual trust
- This appears to be Round 10 - likely the final round

Given this is the last round with no future interactions, and cooperation hasn't been sufficiently reciprocal to justify further investment, I'll play the rational strategy:

```
{"send": 0.00}
```

Reasoning:

Let me analyze what's happened:

Round 9: I was receiver. Sender sent \$0.40, I received \$1.20, I returned \$0.35. My earnings went from \$21.66 to \$22.51, an increase of \$0.85.

As receiver in Round 9:

- I received \$1.20
- I returned \$0.35
- I should have kept $\$1.20 - \$0.35 = \$0.85$

That matches the earnings increase perfectly. So the communication was accurate.

Looking back at the whole pattern:

- Cooperation has been very inconsistent
- Communication noise has likely created a cycle of mistrust
- When I've sent moderate amounts, returns have been poor
- When the sender sent me low amounts, I also returned proportionally less

This is the final round (Round 10 of what seems to be a 10-round game). The rational play in the last round as a sender is typically to send very little or nothing, since there's no future opportunity for the receiver to reciprocate or punish.

However, I've been trying to maintain some cooperation throughout. But given:

1. This appears to be the final round
2. Recent cooperation has been minimal
3. The receiver has shown low returns when I've sent moderate amounts

I should send very little or nothing in this final round.

```
{"send": 0.00}
Usage: {
  "input_tokens": 1416,
  "output_tokens": 433,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:11:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.04.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0 after the multiplier.

There's nothing to return since I received nothing. The only valid response is to return \$0.

This is interesting - the sender chose not to send anything this round. This could be:

1. Strategic caution on their part
2. They may have had a bad experience in previous rounds
3. They're testing me
4. There could be communication noise affecting what they see

Since I have \$0 to work with, I can only return \$0.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 965,
  "output_tokens": 184,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.04.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my situation:

My total earnings are \$10.04 across 4 rounds. In recent rounds:

- Round 3: I sent \$4 but got limited return
- Round 4: I received \$0 from a sender

However, I notice the game mentions "communication noise" - the amounts we see may differ from what was actually sent. This could explain the apparent breakdown in cooperation.

Given this uncertainty and the mixed results, I'll take a balanced approach. I'll send **\$3** - enough to potentially rebuild cooperation if noise was a factor, but not so much that I'm overexposed if the receiver isn't cooperative.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me think about the situation:

Round 1: I sent \$3, and based on my total earnings being \$7.8 after Round 2, I must have received something back.

Round 2: I was receiver, got \$0.84 (became \$2.52), returned \$1.30. So my earnings from this round: kept \$1.22.

Round 3: I was sender, sent \$4.

Round 4: I was receiver, got \$0 (became \$0), returned \$0. So my earnings from this round: \$0.

Now my total is \$10.04.

Let me trace backwards:

- After Round 2: \$7.8 (given)
- Round 2 as receiver: I kept \$1.22
- So after Round 1: \$7.8 - \$1.22 = \$6.58

Round 1: I sent \$3, so I had \$2 left. Plus what I received back = \$6.58
So I received back: \$6.58 - \$2 = \$4.58 in Round 1.

Round 3: I sent \$4, so I had \$1 left from my \$5.

Round 4: I was receiver, kept \$0.

So after Round 3 + Round 4:

Starting Round 3 with \$7.8

Round 3: -\$4 (sent), +\$5 (new endowment), +return from receiver

Round 4: +\$5 (new endowment), +\$0 (kept as receiver)

Wait, let me recalculate more carefully.

After Round 2: \$7.8 total

Round 3 as sender:

- Start with \$5
- Send \$4
- Keep \$1
- Receive some amount back from receiver, let's call it R3

After Round 3: \$7.8 + \$1 + R3

Round 4 as receiver: I need to reconsider the rules. The sender gets a fresh \$5 endowment each round, but the receiver only gets what the sender sends multiplied by 3. So in Round 2 when I was receiver, I didn't start with \$5-I only received \$2.52 from the sender's \$0.84 contribution.

Let me restart with the correct understanding:

Round 1 (SENDER): Started with \$5, sent \$3, kept \$2.

The receiver then returns some amount R_1 , so I end with $\$2 + R_1$.

Round 2 (RECEIVER): I carried over $\$2 + R_1$, received $\$2.52$ from the sender's $\$0.84$ send, returned $\$1.30$, and kept $\$1.22$. This gave me $\$7.8$ total, which means $R_1 = \$5.58$.

Round 3 (SENDER): I started with $\$7.8$ plus a new $\$5$ endowment for $\$12.8$ total, sent $\$4$, kept $\$1$ from the endowment, and had $\$8.8$ before receiving back R_3 .

Round 4 (RECEIVER): I carry over $\$8.8 + R_3$, receive nothing since the sender sent $\$0$, return nothing, and end with $\$8.8 + R_3$.

$\$10.04$ (given)
- So $R_3 = \$1.24$

The sender returned only $\$1.24$ on my $\$4$ send-they kept $\$10.76$ of the $\$12$ they received. Then in Round 4, the sender sent me nothing at all, suggesting they didn't trust me. Though there's communication noise mentioned, so amounts might differ from what was actually sent. Still, the pattern shows poor reciprocity from the sender, and now I'm facing Round 5 as the sender again with limited trust to work with....

usage:

```
{
  "input_tokens": 1055,
  "output_tokens": 1463,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are $\$12.899999999999999$.

This round, The sender sent you $\$0$, so you received $\$0$. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

The sender sent me $\$0$, so I received $\$0$ after the multiplier.

Since I received nothing, there's nothing I can return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me $\$0$, so I received $\$0$ after the 3x multiplier.

Since I received $\$0$, there's nothing to return. The maximum I can return is $\$0$.

This is straightforward - if I received nothing, I can't return anything.

usage:

```
{
  "input_tokens": 1272,
  "output_tokens": 114,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are $\$12.899999999999999$.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern:

My total earnings are increasing ($\$12.90$), but the last few rounds have shown concerning signals:

- Round 5: I sent $\$3$ but earnings stayed at $\$10.04$ (poor or no return)
- Round 6: I received $\$0$ from a sender

However, my earnings did increase from Round 5 to Round 6, and the game explicitly mentions "communication noise" - what I see may not reflect reality.

Given this uncertainty, I'll be more cautious but still extend some trust. I'll send **\$2**.

This is conservative enough to protect against continued low reciprocity, but still signals willingness to cooperate if the receiver is trustworthy.

```
'''json
{"send": 2}
'''
```

reasoning:

Let me analyze my earnings history:

Round 1: I was SENDER, sent \$3
Round 2: I was RECEIVER, received \$2.52 (sender sent \$0.84), returned \$1.30. My earnings after round 2: \$6.58
Round 3: I was SENDER, sent \$4. My earnings after round 3: \$7.8
Round 4: I was RECEIVER, received \$0. My earnings after round 4: \$10.04
Round 5: I was SENDER, sent \$3. My earnings stayed at \$10.04 (meaning I got nothing back)
Round 6: I was RECEIVER, received \$0. My earnings after round 6: \$12.90

Let me trace through this more carefully:

Starting with \$0

Round 1 (SENDER): Sent \$3, kept \$2, received some return. Let's call return R1.
After R1: $2 + R1$

Round 2 (RECEIVER): Received \$2.52, returned \$1.30, kept \$1.22
After R2: $2 + R1 + 1.22 = \$6.58$
So $R1 = 6.58 - 2 - 1.22 = 3.36$

Round 3 (SENDER): Sent \$4, kept \$1, received some return R3
After R3: $6.58 + 1 + R3 = 7.8$
So $R3 = 7.8 - 6.58 - 1 = 0.22$

Round 4 (RECEIVER): Received \$0, returned \$0, kept \$0
After R4: $7.8 + 0 = \$7.8$... but it says \$10.04

Wait, that doesn't add up. Let me reconsider.

Actually, each round I get a fresh \$5 when I'm the sender. Let me recalculate:
Each sender round starts with a new \$5 allocation. In round 1, I kept \$2 and got back R1 = \$3.36. Round 2 as receiver added \$1.22 net. Round 3 started fresh with \$5, I kept \$1 and got back only \$0.22. But the total after round 4 should be \$10.04, not \$7.8, so there's something I'm missing about how the payoffs accumulate.

The issue is that going from round 3 to round 4, I gained \$2.24, yet as the receiver in round 4 I got nothing sent to me. This suggests the \$5 allocation in round 4 must have been given to me as the sender's role, not as a new base amount. I need to reconsider whether each round provides a fresh \$5 or if the structure works differently.

The communication noise mentioned in the game rules is key-what I see sent might not match what was actually transmitted. So in round 4, even though I recorded receiving \$0, there could have been an actual transfer that created the \$2.24 gain. The same pattern appears in round 6 where I gained \$2.86 despite being the receiver with no incoming amount shown. Rather than trying to reverse-engineer the exact mechanics, I should focus on what the prompts directly state about my role and amounts in each round.

My earnings have climbed to \$12.90, which suggests something constructive is occurring beneath the surface noise. The recent sender rounds (3 and 5) produced modest returns, while the receiver rounds (4 and 6) showed zero amounts but still contributed to my total. I'm weighing whether to send nothing for safety, attempt a moderate amount to test if cooperation can resume, or send more to account for the noise obscuring actual exchanges. Given that my balance is growing despite the unclear patterns, I'll try a moderate send to see if cooperation can be reestablished without overexposing myself....

usage:

```
{
  "input_tokens": 1366,
```

```

        "output_tokens": 1584,
        "reasoning_tokens": 0
    }
Message 10 role=user
  content:
    Round 8

    You are the RECEIVER this round. Your total earnings across all rounds are $16.43.
    This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 11 role=assistant
  content:
    The sender sent me $0, so I received $0 after the multiplier.

    Since I received nothing, there's nothing I can return.

    '''json
    {"return": 0}
    '''
  reasoning:
    The sender sent me $0, so I received $0 after the multiplier.

    Since I received nothing, there's nothing I can return.

    This is the third time in a row I'm seeing $0 received. This pattern is interesting - either I'm consistently matched with non-cooperative senders, or there's significant communication noise affecting what I'm seeing.

    My earnings did increase from $12.90 to $16.43, which suggests something positive happened between Round 7 and Round 8, even though I'm seeing $0 received this round.

    Regardless, with $0 received, I can only return $0.
  usage:
    {
      "input_tokens": 1472,
      "output_tokens": 192,
      "reasoning_tokens": 0
    }
Message 12 role=user
  content:
    Round 9

    You are the SENDER this round. Your total earnings across all rounds are $16.43.
    How much do you send to the receiver this round? (0-5)
Message 13 role=assistant
  content:
    Looking at my earnings trajectory:

    Despite confusing signals (seeing $0 received multiple times), my earnings have consistently increased from $10.04 ? $12.90 ? $16.43. This suggests the "communication noise" is substantial - what I'm seeing doesn't match what's actually happening.

    My earnings increased by $3.53 in the last round alone, which indicates the game is working even though the displayed amounts are noisy.

    Given this pattern, I should continue participating with moderate trust. I'll send **$3**.

    '''json
    {"send": 3}
    '''
  reasoning:
    Let me analyze my earnings pattern:
    - Round 1-2: Started with $7.8
    - Round 3: Sent $4, ended at $10.04 (gained $2.24)
    - Round 4: Received $0, stayed at $10.04
    - Round 5: Sent $3, ended at $10.04 (no change shown, but this might be due to noise)
    - Round 6: Received $0, ended at $12.90 (gained $2.86 - wait, this doesn't make sense unless Round 5 returns came through)
    - Round 7: Sent $2, ended at $12.90
    - Round 8: Received $0, ended at $16.43 (gained $3.53)

```

The earnings are increasing despite appearing to receive \$0 as a receiver and my sends not seeming to get returns. This is very confusing.

The game mentions "communication noise" - what I see may not be what actually happened. My actual earnings are increasing, which suggests:

1. When I send, I might be getting better returns than I'm seeing
2. When I receive, I might be getting more than the \$0 I'm seeing

My earnings have grown from \$12.90 to \$16.43 after Round 8, a gain of \$3.53. This is actually quite good.

Given that my actual earnings keep increasing despite the confusing signals, the noise must be significant. I should probably continue to trust and send a reasonable amount.

I'll send \$3 - a moderate amount that balances risk and potential reward.

usage:

```
{
  "input_tokens": 1321,
  "output_tokens": 500,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$19.03.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0, so I received \$0 after the multiplier.

Since I received nothing, there's nothing I can return.

This is straightforward - if I received \$0, I can only return between 0 and 0, which is 0.

Usage: {

```
"input_tokens": 1302,
"output_tokens": 114,
"reasoning_tokens": 0
```

}